

# The Dimension of Nonterminating Resampling Computations

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## Abstract

A randomized algorithm may terminate almost surely even though exceptional random tapes make it run forever. This paper studies their survival probability, individual Kolmogorov complexity, and Hausdorff dimension. The main result develops source-power bounds for matrix-commutative local repair. Whenever the powered repair kernels commute, a parameter-wise trace/Shearer inequality controls survival and Hausdorff dimension and, under computability, individual complexity. The resulting bounds extend to history-dependent weak randomness, uniformly over deterministic nonanticipating repair rules. For  $k$ -SAT with clause-overlap graph  $D$ , conditional min-entropy exceeding  $\log_2 \kappa_D$  per redraw yields an exponential tail, where  $\kappa_D$  is the trace-growth rate. Tree and clique constructions establish sharpness for the maximum-degree dimension bound and the graph-specific one-step threshold, respectively, in their stated regimes; the method also applies to rainbow matching and localizes one infinite run to the clauses repaired infinitely often. A complementary backward likelihood identity gives tail and coding bounds for individual runs. Exact reductions and hardness results identify tractable regimes and computational obstacles for survival pressure and nontermination dimension.

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# 1 Introduction

A randomized algorithm is deterministic once its random tape is fixed. Although the algorithm may terminate with probability one, some tapes may make it run forever. Let  $\mathbb{P}$  denote the source law of the tape and put

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where  $\tau$  is the number of rejection or repair steps before stopping. The tail  $\mathbb{P}\{\tau \geq T\}$  measures how often the computation survives  $T$  steps. When the following extended limit exists, the *survival exponent* (or base-two escape rate) is

$$\alpha = - \lim_{T \rightarrow \infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

This terminology is standard for exponential escape from a surviving region (Keller and Liverani, 2009). This distributional quantity does not describe the exceptional tapes themselves. For a computable process with a computable tape law  $P$ ,  $\mathcal{N}$  is effectively closed because termination, when it occurs, is determined by a finite tape prefix. Hence a  $P$ -null  $\mathcal{N}$  contains no  $P$ -Martin-Löf-random tape. It may nevertheless have positive or even full Hausdorff dimension and contain tapes of maximal effective strong dimension. Effective dimension measures the algorithmic randomness of one fixed tape; source-relative Hausdorff dimension measures the size of the whole exceptional set.

The Lovász local lemma has a different scope. It gives conditions under which all bad events can be avoided, and its algorithmic forms give termination or tail guarantees from event probabilities and dependency structure (Erdős and Lovász, 1975; Moser and Tardos, 2010; Haeupler and Harris, 2017). The present paper fixes the resampling computation and asks for its exact survival decay and for the complexity and geometry of its exceptional tapes. These are questions about a specified process, not a stronger universal local-lemma criterion. In a finite full-support variable model, mere existence of a flawless assignment already implies some exponential tail (Proposition 4.7); the local lemma is valuable because it supplies efficient, locally verifiable quantitative bounds. The present paper asks for the rate and the tapes hidden beneath that qualitative boundary.

## 1.1 Pointwise complexity and localization

In statistical learning,  $\hat{h}(S)$  depends on the training sample  $S$ , so a bound for one fixed hypothesis cannot simply be evaluated at the learned one. Uniform and localized empirical-process methods instead control the data-dependent region in which the estimator lies (Bartlett et al., 2005; Koltchinskii, 2006; Xu and Zeevi, 2025). A resampling computation has the analogous dependence between a random tape and the repair history it generates:  $\{\tau \geq T\}$  restricts tape space to the nested family of live prefixes selected by the computation. Recent work localizes complexity at one trained neural network or one index of a random field (Li and Xu, 2026; Xu, 2026b); here the pointwise object is one random tape. The later search for a smaller state preserving specified quantities also parallels decision-sufficient state compression (Xu, 2026a). These parallels motivate the questions but are not reductions: local Rademacher complexity, Kolmogorov complexity, and weighted cylinder sums are distinct objects, and no learning-theoretic generalization bound is claimed. Corollary 3.7 nevertheless makes one localization principle exact: the asymptotic complexity of an infinite  $k$ -SAT run depends only on the clauses repaired infinitely often.

For a computable full-support source  $P = \pi^{\mathbb{N}}$ , let  $\omega_{1:n}$  denote the first  $n$  source symbols (or blocks). For a finite word  $w$ , let  $[w]$  be the cylinder of tapes beginning with  $w$ , and write  $P[w] = P([w])$ . Its self-information is

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}].$$

Let  $K(w)$  be the length of the shortest program that prints  $w$  on a fixed universal prefix-free, or self-delimiting, machine (Li and Vitányi, 2019). More generally,  $K(w \mid z)$  is the corresponding program length when  $z$  is supplied as auxiliary input. The effective strong  $P$ -dimension of one tape is

$$\text{Dim}_{\text{eff},P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \quad (1)$$

Replacing the limsup by a liminf defines the effective  $P$ -dimension  $\text{dim}_{\text{eff},P}(\omega)$ . For a fair binary source, write

$$\text{Dim}_{\text{eff}}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{n}.$$

These are the standard constructive dimensions relative to a source measure (Athreya et al., 2007; Lutz, 2011); the definitions themselves are not new. A value near one means that the tape retains nearly the full algorithmic information of the source, while a smaller value means that continued execution has imposed additional structure.

For a set  $E$  of tapes,  $\text{dim}_H^P E$  denotes its Billingsley, or source-relative Hausdorff, dimension: in the Hausdorff covering construction, the cylinder  $[w]$  has diameter  $P[w]$  (Billingsley, 1965; Lutz, 2011). Under a uniform  $q$ -ary source, this is ordinary Hausdorff dimension in the symbolic metric  $d(\omega, \eta) = q^{-k}$ , where  $k$  is the length of the longest common prefix. Section 2.1 gives the corresponding source tree and metric when the amount of tape consumed is adaptive.

Dimension also has an operational interpretation. The mass-distribution principle and Frostman's lemma imply, for every closed  $E$  in this symbolic space,

$$\text{dim}_H^P E = \sup \left\{ s : \begin{array}{l} \text{there are a probability law } \nu \text{ with } \nu(E) = 1 \text{ and } D < \infty \\ \text{such that } \nu[w] \leq DP[w]^s \text{ for every finite word } w \end{array} \right\}. \quad (2)$$

This is the classical Frostman characterization specialized to the source metric (Falconer, 2014), not a new dimension theorem. It explains why source powers are algorithmically useful: the dimension of the exceptional set is exactly the largest global min-entropy rate that a tape law supported on that set can retain. A conditional min-entropy bound at every repair is a strong, directly checkable way to obtain the cylinder inequality in (2).

## 1.2 Two complementary principles

The paper uses two principles with different quantifiers. The first is pathwise. Comparing one finite run with any subprobability law  $Q_T$  on feasible repair histories gives the information surplus

$$\Delta_T = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(A_1, \dots, A_T)}.$$

Reading the run backward shows that this is an exact log-likelihood ratio. Consequently  $\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}$ ; under computability, the same backward law describes each surviving prefix using  $\Delta_T$  fewer bits than its source self-information, up to an additive constant. This identity requires no local-lemma hypothesis.

The second principle is setwise. For the prefix-free family  $\mathcal{L}_T$  of consumed tape prefixes that remain live for  $T$  repairs, define

$$Z_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s.$$

This is the Rényi partition sum of the live cylinders, whose exponential growth is pressure (Rényi, 1961; Bowen, 1975). At  $s = 1$  it is the survival probability; its critical power gives Hausdorff

dimension. For a finite-state computation it is governed by a nonnegative operator  $B_s$ . If a statistic preserves the total  $p(e)^s$ -weight of transitions between its classes for every  $s$ , a reduced matrix  $R_s$  preserves  $Z_T(s)$  exactly. Thus  $\rho(R_1)$  gives the survival rate and the solution of  $\rho(R_\delta) = 1$  gives the dimension.

These principles are complementary. The likelihood identity concerns one realized run and needs no finite-state description. Pressure sums over all surviving runs and gives exact asymptotic answers when their weighted growth has a tractable representation. Neither is a consequence of the other.

### 1.3 Main contributions

**Source powers and matrix-commutative repair.** The central result controls the entire source-power family of a local repair process, rather than only its ordinary survival probability. Lemma 3.2 bounds every full repair-history DAG by a product of powered transition matrices. Summing these bounds, Theorem 3.3 gives a parameter-wise trace/Shearer inequality for  $Z_T^\Lambda(s)$ , uniformly over every deterministic nonanticipating selector. The proof works directly with the powered matrices  $K_{f,s}$ . It requires neither recovery of an action label from a state transition nor an action-level swapping construction. At  $s = 1$  the result controls termination; for  $s < 1$  it bounds Hausdorff dimension, pointwise effective strong dimension under computability, and history-dependent weak sources. Product-preserving action diamonds remain sufficient, while uniform fixed-arity oracles need only ordinary matrix commutation.

**Weak-source robustness and sharp dimension bounds.** The main operational consequence concerns bounded-dependence  $k$ -SAT. Put

$$A_d = 1 + \frac{d^d}{(d-1)^{d-1}}.$$

Theorem 3.6 proves exponential termination above  $\log_2 \kappa_D$  bits of conditional min-entropy per repaired clause, where  $\kappa_D$  is the trace-growth rate of the actual overlap graph. For maximum degree  $d$ ,  $\kappa_D \leq c_d = d^d/(d-1)^{d-1}$ . For  $2 \leq d \leq k$ , tree formulas with variable occurrence two asymptotically attain  $c_d$  for dimension and global source domination, including at the critical power (Theorem 3.12); when  $d+1 \leq 2^{k-1}$ , a clique construction is exact for the one-step threshold on its graph. When  $A_d \leq 2^{k-1}$ , the explicit local bound in Theorem 3.8 yields  $O((n+m/d) \log(d+1))$  expected fresh fair bits, and Santha–Vazirani tapes require no extractor. For a computable process, Corollary 3.7 bounds one tape’s complexity by the trace growth of the clauses repaired infinitely often. Rainbow perfect matching gives a non-product-space application.

**Pathwise identity, exact regimes, and computational limits.** Theorem 2.1 is an exact finite-run likelihood and KL identity; it gives a universal tail bound and a pointwise coding bound without a local-lemma assumption. Theorem 4.3 computes the survival exponent and both dimensions for finite extremal partial rejection, while Theorem 4.9 gives a separate exact all-power reduction for nonextremal disagreement repair on every connected graph. The strict survival and dimension tests are NP-complete for fixed-priority three-variable repair of dependency degree at most nine (Theorem 4.8); stronger exact and approximation questions are #P-hard or NP-hard in the succinct models studied later. Finally, matched whole-block systems show that the event–variable graph, event marginals, and all pairwise intersections can omit higher-order information that changes both survival and dimension.

## 1.4 Relation to earlier work and scope

Reverse reconstruction is an established entropy-compression device (Moser, 2009; Moser and Tardos, 2010). In particular, Messner and Thierauf (2012) reconstruct the uniform random bits used by Moser’s satisfiability search from the final assignment and a compressed repair log, and then apply Kolmogorov complexity. In the broader framework of focused stochastic search, Achlioptas et al. (2019, Theorem 5) give sharp product-form probabilities for feasible flaw sequences under atomicity and regeneration. The backward theorem here does not claim reconstruction as new. Its contribution is an exact likelihood formulation for arbitrary finite product sources and non-atomic variable-model events, the freedom to choose  $Q_T$ , and the resulting KL slack and pointwise coding bound.

For extremal partial rejection, confluence, Cartier–Foata traces, and the exact probability-weighted transcript series are known (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024). Abbes and Mairesse (2015) characterize Bernoulli measures on irreducible infinite trace boundaries through Möbius vanishing and positivity and realize them as Markov chains on Cartier–Foata cliques. Thus neither trace enumeration nor the existence of a critical trace-boundary law is claimed as new. The additional step is to power each consumed source value before aggregation, replacing  $p_i^s$  by its within-event Rényi mass, and to identify the resulting critical power with the source-relative Hausdorff and maximal effective strong dimensions of the consumed tapes. For reducible dependency graphs, the analysis is componentwise.

The finite-state connection among weighted growth, Perron eigenvectors, and Hausdorff dimension is classical (Parry, 1964; Bowen, 1975; Mauldin and Williams, 1988; Simpson, 2015). At  $s = 1$  the reduction below is ordinary Markov lumpability, closely related to probabilistic bisimulation (Kemeny and Snell, 1960; Larsen and Skou, 1991); its all-power form is naturally compared with weighted stochastic automata (Carlyle and Paz, 1971). The continuation lower bounds are Myhill–Nerode-style distinctions. Probabilistic-program verification usually studies reachability, almost-sure termination, or expected time (Etessami and Yannakakis, 2009; Fioriti and Hermanns, 2015); the present question concerns the individual complexity and Hausdorff geometry of the exceptional tapes of a fixed computation.

Action-level commutativity for resampling oracles was developed by Kolmogorov (2018); the matrix formulation of Harris et al. (2025) is strictly broader. The central theorem below works directly with these matrices at every source power  $s$  for which the powered matrices commute. Product-preserving action diamonds imply this condition for every  $s$ . So does ordinary matrix commutation for uniform fixed-arity oracles. Without such additional structure, commutation at  $s = 1$  need not survive powering. The ordinary trace/Shearer analysis at  $s = 1$  is part of the established commutative-LLL theory. The exact-prefix matrix argument below lifts that analysis to the whole source-power family and derives the weak-source, Hausdorff-dimension, and pointwise-complexity consequences from the same bound.

The dependent-source corollary is also narrower than general weak-randomness simulation. Chor–Goldreich sources impose conditional min-entropy on successive blocks (Chor and Goldreich, 1988); general randomized algorithms can be simulated from related sources using online condensation (Doron et al., 2023). Here the original repair algorithm is run without an extractor. The guarantee is instance-specific: a local source-power inequality gives its tail and nontermination dimension directly. The Santha–Vazirani specialization uses the classical bitwise source model of Santha and Vazirani (1986). A different line of work reduces the total randomness of a parallel Moser–Tardos variant by reusing bits at distant repairs on dependency graphs of subexponential growth (Csóka et al., 2022). The result here is incomparable: it uses fresh bits at each repair and does not obtain constant total randomness, but it requires only bounded dependency, permits adaptive injective redraw maps and arbitrary nonanticipating selectors, and tolerates history-dependent weak sources

directly.

The powered theorem is a selector-uniform upper-bound principle, not a new universal LLL threshold and not an equality theorem for every repair process. Exactness is proved only for the extremal and disagreement-repair classes analyzed later. The hardness results rule out unrestricted exact computation from succinct input, but leave open a positive compilation theorem for a natural bounded-width class of overlapping repairs. Rejection sampling is used only as a one-state calibration. Sections 2 and 3 develop the pathwise and setwise principles; Sections 4 and 5 give exact regimes and their informational and computational limits. All proofs are collected in four appendices.

## 2 Pathwise analysis: a backward likelihood identity

Let  $X_1, \dots, X_n$  be independent variables, with  $X_i$  taking values in a finite set  $\mathcal{X}_i$ , and with product law  $\pi = \bigotimes_i \pi_i$ . For  $J \subseteq \{1, \dots, n\}$ , write  $\pi_J = \bigotimes_{i \in J} \pi_i$  for the marginal product law on the coordinates in  $J$ . Each nonempty bad event  $A$  is determined by the coordinates  $v(A)$  and has probability  $p_A > 0$ . Its *marginal surprisal* is  $\log_2(1/p_A)$ . Fix a deterministic rule for selecting a currently true bad event.

List the random choices in the order in which the algorithm uses them. First draw  $S_0 \sim \pi$ . Whenever the current assignment contains a bad event, select  $A_t$ , draw  $Z_t \sim \pi_{v(A_t)}$  independently of the past, and write  $Z_t$  on  $v(A_t)$ . Let  $S_t$  denote the resulting assignment after the  $t$ th repair. Let  $E_T$  be the event that at least  $T$  resamplings occur, and put

$$\Xi_T = (S_0, Z_1, \dots, Z_T).$$

### 2.1 The consumed-tape space

The tape is the rooted tree of source choices actually queried by the computation. A finite node is a feasible transcript

$$w = (s_0, z_1, \dots, z_t).$$

The root branches to  $s_0$  with probability  $\pi(s_0)$ . If  $w$  is live and the rule next selects  $A_{t+1}$ , its children are indexed by  $z \in \prod_{i \in v(A_{t+1})} \mathcal{X}_i$ , with edge probability  $\pi_{v(A_{t+1})}(z)$ . Thus the cylinder rooted at  $w$  has mass

$$P_{\text{tape}}[w] = \pi(s_0) \prod_{j=1}^t \pi_{v(A_j)}(z_j). \quad (3)$$

Terminal transcripts are leaves, and infinite branches are precisely the nonterminating consumed tapes. For distinct infinite branches  $\omega$  and  $\eta$ , let  $\omega \wedge \eta$  be their longest common consumed prefix and set

$$d_P(\omega, \eta) = P_{\text{tape}}[\omega \wedge \eta]. \quad (4)$$

Equivalently, source-relative Hausdorff dimension uses the cylinder gauge  $P_{\text{tape}}[w]$ . All dimension results below assume that these masses tend to zero along every infinite branch.

Fix computable enumerations of the finite outcome alphabets. They give a canonical self-delimiting encoding of every transcript; the selected scope is recovered from the preceding transcript and the deterministic rule. The notation  $K(w)$  refers to this encoding. The dimensions are unchanged under a computable prefix-tree isomorphism  $\Psi$  with computable inverse and bounded distortion

$$C^{-1} P_{\text{tape}}[w] \leq P'_{\text{tape}}[\Psi(w)] \leq C P_{\text{tape}}[w]$$

for some fixed  $C$ : the two self-informations and the two prefix complexities then differ by  $O(1)$ . No invariance under arbitrary recodings is claimed.

A pre-sampled resampling table is only a coupling device. Entries not read by the realized computation are excluded: adjoining independent unused coordinates can change a tape dimension without changing either the computation or its stopping time.

Let  $P_{\text{tape}}$  denote this sequential source law, and abbreviate  $P_{\text{tape}}(w) = P_{\text{tape}}[w]$ . Thus  $E_T = \{\tau \geq T\}$ . The minimal consumed prefixes that realize  $T$  repairs form a prefix-free family, so their cylinders are disjoint and their probabilities sum to  $\mathbb{P}(E_T)$ .

Let  $Y_t = S_{t-1}|_{v(A_t)}$  be the configuration overwritten at time  $t$ , and let  $L_T = (A_1, \dots, A_T)$ . Write  $\pi_{v(A)}(\cdot | A)$  for the marginal law  $\pi_{v(A)}$  conditioned on the local configurations satisfying  $A$ . Coordinatewise, the entries of the used tape are partitioned into the final assignment and the overwritten configurations. Therefore

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T p_{A_t} \pi_{v(A_t)}(Y_t | A_t). \quad (5)$$

Let  $Q_T$  be a subprobability law on repair histories  $\ell = (A_1, \dots, A_T)$ , positive on every feasible history of length  $T$ . Thus  $-\log_2 Q_T(\ell)$  is its ideal code length. Define the backward subprobability law on complete records by

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^T \pi_{v(A_t)}(y_t | A_t). \quad (6)$$

The map

$$\Xi_T \mapsto (S_T, L_T, Y_{1:T})$$

is injective. Indeed, reverse the repairs from  $T$  to 1: the current values on  $v(A_t)$  recover  $Z_t$ , and restoring  $Y_t$  recovers the preceding assignment.

Reverse reconstruction itself is an established entropy-compression device. In particular, [Messner and Thierauf \(2012\)](#) reconstruct the uniform random bits used by Moser's satisfiability search from its repair log and final assignment, and then apply Kolmogorov complexity. The result below adds an exact likelihood formulation for nonuniform sources and non-atomic events, an arbitrary comparison law  $Q_T$ , and a common tail, KL, and coding interpretation.

**Theorem 2.1** (Backward likelihood identity). *Every execution in  $E_T$  satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \quad (7)$$

Call the right side  $\Delta_T$  on  $E_T$  and set  $\Delta_T = -\infty$  otherwise. Then, for every  $u \geq 0$ ,

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \quad (8)$$

If the instance and the family  $(Q_T)_{T \geq 1}$  are uniformly computable, then

$$-\log_2 P_{\text{tape}}(\Xi_T) - K(\Xi_T | T, \text{instance}) \geq \Delta_T - O(1). \quad (9)$$

Here instance denotes a fixed effective description of the source laws, bad events, and deterministic selection rule. Uniform computability means that one algorithm can approximate  $Q_T(\ell)$  to any requested precision from  $T$  and  $\ell$ , with the instance data encoded effectively in the same sense.

For atomic regenerating focused stochastic search, Achlioptas et al. (2019, Theorem 5) obtain sharp product-form probabilities for feasible flow sequences. The present record retains the over-written local configurations, so the identity also applies to non-atomic variable-model events; the arbitrary law  $Q_T$  then exposes both the exact mismatch in (10) below and the coding consequence (9).

For use in the exact slack formula below, let  $\mathcal{W}_T$  be the prefix-free set of minimal tape prefixes producing at least  $T$  repairs and write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)), \quad \mu_T(w) = G_T(R_T(w)).$$

Backward reconstruction makes  $R_T$  injective. Since  $G_T$  has total mass at most one,  $\mu_T$  is a subprobability mass on  $\mathcal{W}_T$ .

The proof is given in Appendix B.

The likelihood identity also identifies the exact slack after conditioning on a long run. Let  $p_T = P_{\text{tape}}(E_T) > 0$ , let  $m_T = \sum_{w \in \mathcal{W}_T} \mu_T(w) > 0$ , and define probability laws on  $\mathcal{W}_T$  by

$$P_T(w) = \frac{P_{\text{tape}}(w)}{p_T}, \quad \nu_T(w) = \frac{\mu_T(w)}{m_T}.$$

Write  $D_2$  for relative entropy in bits and  $H_2$  for Shannon entropy in bits.

**Corollary 2.2** (Exact slack and the optimal history law). *For every fixed comparison law  $Q_T$ ,*

$$-\log_2 p_T = \mathbb{E}_{P_T} \Delta_T + D_2(P_T \| \nu_T) - \log_2 m_T. \quad (10)$$

Moreover, if  $J_T = \sum_{t=1}^T \log_2(1/p_{A_t})$ , then

$$\sup_{Q_T} \mathbb{E}[\Delta_T \mid E_T] = \mathbb{E}[J_T \mid E_T] - H_2(L_T \mid E_T), \quad (11)$$

where the supremum is over subprobability laws positive on feasible histories. It is attained by  $Q_T(\ell) = \mathbb{P}\{L_T = \ell \mid E_T\}$  on the conditional support.

The proof is given in Appendix B.

Thus the difference between the escape information  $-\log_2 p_T$  and the mean surplus consists exactly of a mismatch term and the unused backward mass. The theorem also gives the normalized bound

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \geq a\right\} \leq 2^{-aT} \quad (T \geq 1, a \geq 0). \quad (12)$$

**Corollary 2.3** (Sharpness of the surplus tail). *Neither the constant nor the exponent in (12) can be improved universally. For any  $a > 0$ , a one-event rejection sampler with event probability  $p = 2^{-a}$  and its unique repair history assigned probability one satisfies, for every  $T$ ,*

$$\mathbb{P}\left\{\frac{\Delta_T}{T} \geq a\right\} = \mathbb{P}(E_T) = p^T = 2^{-aT}. \quad (13)$$

**Large-deviation interpretation.** No large-deviation principle is assumed in the paper. If  $\Delta_T/T$  is known separately to satisfy one with a base-two rate function  $J$ , and the principle identifies the exact upper-tail rate at  $a$ , then

$$\inf_{x \geq a} J(x) \geq a. \quad (14)$$

The one-event example shows that equality is possible. For ordinary rejection sampling with rejection probability  $p$ , survival means that  $T$  independent Bernoulli variables all equal one, so its rate is the endpoint relative entropy

$$D_2(1\|p) = \log_2 \frac{1}{p}.$$

For the finite-state model in Appendix A, the corresponding escape rate is  $-\log_2 \rho(B_1)$  (Dembo and Zeitouni, 2010; Keller and Liverani, 2009).

The two terms in (7) have different origins. The first sums the marginal surprisals of the events that were repaired; the second is the ideal code length of their ordered history. The first sum may count statistically dependent information more than once. Neither term alone gives a bound. Their difference does, because it is a likelihood ratio.

For example, color the vertices of a graph independently with  $q$  colors and repeatedly recolor the endpoints of the first monochromatic edge. A fixed edge is monochromatic with probability  $1/q$ , so each repair contributes  $\log_2 q$  bits to the first term. The second term pays for the repaired edges and their order. The comparison is therefore between information ruled out by the violations and information used by the algorithm's history.

**Corollary 2.4** (A tail bound and a pointwise dimension bound). *Suppose every  $T$ -repair execution satisfies*

$$\sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} \geq \beta T - C_0, \quad -\log_2 Q_T(L_T) \leq \gamma T + C_1,$$

where  $\beta > \gamma$ . Then

$$\mathbb{P}(E_T) \leq 2^{C_0 + C_1 - (\beta - \gamma)T}. \quad (15)$$

For the pointwise conclusion, assume in addition that

- (i) the instance and the family  $(Q_T)_{T \geq 1}$  are uniformly computable;
- (ii) the computation reads a fixed fair binary tape sequentially, the initial draw consumes  $c_0 = O(1)$  bits, and every repair consumes exactly  $c$  bits, so that  $\Xi_T$  is uniformly computably interconvertible with  $\omega_{1:c_0+cT}$ .

Then every nonterminating tape satisfies

$$\text{Dim}_{\text{eff}}(\omega) \leq \max \left\{ 0, 1 - \frac{\beta - \gamma}{c} \right\}. \quad (16)$$

The proof is given in Appendix B.

*Remark 2.5* (What the record must contain). For a recursive Moser-style implementation,  $L_T$  must be enlarged to include the recursion tree or return structure unless those data are recoverable by a separate prefix-free decoding rule. In Theorem 2.1, everything needed for reversal contributes to the probability  $Q_T(L_T)$ .

### 3 Setwise analysis: powered matrix-commutative repair

This section bounds the complete source-power sum of an overlapping repair process without constructing its full assignment chain. The input is a family of powered transition matrices and local scalar bounds on their growth. The conclusion is uniform over the order in which present flaws are selected.

Let  $\Omega$  be a finite state space and let  $F_\sigma \subseteq F$  be the flaws present at  $\sigma$ . Addressing  $f \in F_\sigma$  chooses an action  $a \in \mathcal{A}_f(\sigma)$  with probability  $\rho_f(a \mid \sigma) > 0$  and moves to  $\Phi_f(\sigma, a)$ . The action sets are finite and satisfy

$$\sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma) = 1.$$

Distinct actions may lead to the same terminal state. A deterministic nonanticipating selector may use the complete observed past, including past action labels, but not future randomness.

Let  $\sim$  be a symmetric relation on distinct flaws and define the inclusive neighborhood

$$\Gamma(f) = \{f\} \cup \{g \in F : g \sim f\}.$$

In local-repair applications,  $\sim$  is also chosen to satisfy the potential-causality condition

$$F_{\Phi_f(\sigma, a)} \subseteq (F_\sigma \setminus \{f\}) \cup \Gamma(f). \quad (17)$$

This is Kolmogorov's convention after adding a self-loop at every flaw. The matrix theorem itself uses only powered commutation; potential causality explains why  $D$  is a local dependency graph. Write  $\text{Ind}(S)$  for the subsets of  $S$  whose distinct elements are pairwise nonadjacent.

For  $s > 0$ , define the powered action kernel

$$K_{f,s}(\sigma, \tau) = \mathbf{1}\{f \in F_\sigma\} \sum_{\substack{a \in \mathcal{A}_f(\sigma) \\ \Phi_f(\sigma, a) = \tau}} \rho_f(a \mid \sigma)^s. \quad (18)$$

The kernel aggregates actions with the same endpoints, but for  $s \neq 1$  its entries retain the powered probabilities of the individual actions and need not be determined by the ordinary transition matrix  $K_{f,1}$ . The process is *s-commutative* if

$$K_{f,s}K_{g,s} = K_{g,s}K_{f,s} \quad (f \neq g, f \not\sim g). \quad (19)$$

Two familiar conditions imply this matrix identity. First, suppose that for nonadjacent  $f, g$ , action-labelled paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

are in product-preserving bijection with paths

$$\sigma \xrightarrow{g,b'} \sigma'' \xrightarrow{f,a'} \tau,$$

so that

$$\rho_f(a \mid \sigma) \rho_g(b \mid \sigma') = \rho_g(b' \mid \sigma) \rho_f(a' \mid \sigma''). \quad (20)$$

Second, suppose that each enabled instance of flaw  $f$  has the same number  $M_f$  of equiprobable actions.

**Lemma 3.1** (Two routes to powered commutation). *The product-preserving diamond property implies (19) for every  $s > 0$ . Alternatively, if each enabled  $f$  has exactly  $M_f$  actions of probability  $1/M_f$ , then*

$$K_{f,s} = M_f^{1-s} K_{f,1}. \quad (21)$$

Consequently ordinary matrix commutation at  $s = 1$  implies (19) for every  $s > 0$  in this fixed-arity case. Different action labels may lead to the same terminal state.

The proof is in Appendix B.

Fix an initial law  $u$  on  $\Omega$ . The initial source symbol is taken to be the initial state, and its powered mass vector is

$$b_s(\sigma) = u(\sigma)^s.$$

If instead an initial symbol  $\alpha$  of probability  $q(\alpha)$  produces the state  $\iota(\alpha)$ , replace this by

$$b_s(\sigma) = \sum_{\alpha: \iota(\alpha)=\sigma} q(\alpha)^s.$$

One later source symbol is one complete oracle action. Let  $\mathcal{L}_T^\Lambda$  be the prefix-free family of minimal tape prefixes that complete  $T$  repairs under selector  $\Lambda$ , and set

$$Z_T^\Lambda(s) = \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s, \quad (22)$$

where  $P_{\text{tape}}[w]$  includes the initial source symbol. With this convention,  $Z_T^\Lambda(1) = \mathbb{P}\{\tau \geq T\}$  when  $\tau$  is the number of completed repairs.

Write  $f \simeq g$  when  $f = g$  or  $f \sim g$ . For a flaw word  $\ell = (f_1, \dots, f_T)$ , its *full history DAG* has vertex set  $[T]$  and an edge  $i \rightarrow j$  exactly when  $i < j$  and  $f_i \simeq f_j$ . For such a DAG  $H$ , put

$$K_{\emptyset, s} = I, \quad K_{H, s} = K_{L(x), s} K_{H-x, s},$$

where  $x$  is any source and  $L(x)$  is its label. Under (19), this definition is independent of the chosen source.

Two full history DAGs are identified when they are related by a label-preserving directed-graph isomorphism.

Choose a strictly positive probability vector  $v_s$  on  $\Omega$  and define

$$\lambda_f(s) = \max_{\tau \in \Omega} \frac{\sum_{\sigma \in \Omega} v_s(\sigma) K_{f, s}(\sigma, \tau)}{v_s(\tau)}, \quad (23)$$

$$\gamma_s = \max_{\sigma \in \Omega} \frac{b_s(\sigma)}{v_s(\sigma)}. \quad (24)$$

Let  $D$  be the graph on  $F$  with edges  $f \sim g$  and let  $\mathcal{M}(D)$  be the trace monoid in which nonadjacent flaws commute. For a trace  $h$ , write  $N_f(h)$  for the multiplicity of  $f$  and  $|h| = \sum_f N_f(h)$ . Define

$$a_T(s) = \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_{f \in F} \lambda_f(s)^{N_f(h)}, \quad (25)$$

$$P_D(\mathbf{z}) = \sum_{I \in \text{Ind}(F)} (-1)^{|I|} \prod_{f \in I} z_f. \quad (26)$$

The open Shearer region consists of the vectors  $\mathbf{z} \geq 0$  for which  $P_D(\mathbf{y}) > 0$  for every  $0 \leq \mathbf{y} \leq \mathbf{z}$ . The Cartier–Foata identity gives

$$\sum_{T \geq 0} a_T(s) z^T = \frac{1}{P_D(\mathbf{z} \boldsymbol{\lambda}(s))} \quad (27)$$

as a formal power series, where  $\boldsymbol{\lambda}(s) = (\lambda_f(s))_{f \in F}$ .

At  $s = 1$ , [Harris et al. \(2025, Lemma 3.5\)](#) bound the probability that a prescribed history DAG appears during a run. Their induction allows the current repair to be absent from that DAG and uses substochasticity in this case. Here  $H$  records the entire next  $T$ -repair prefix, so the current repair must label a source of  $H$ . The omission case disappears, allowing the powered matrices to have row sums greater than one when  $s < 1$ .

**Lemma 3.2** (Exact-prefix matrix bound). *Fix  $s > 0$  for which (19) holds. For a full history DAG  $H$  with  $T$  vertices, let  $\mathcal{L}_T^\Lambda(H)$  be the live prefixes whose first  $T$  repair labels induce  $H$ . Every deterministic nonanticipating selector satisfies*

$$\sum_{w \in \mathcal{L}_T^\Lambda(H)} P_{\text{tape}}[w]^s \leq b_s^\top K_{H,s} \mathbf{1}. \quad (28)$$

*No injectivity of the action-to-state map is required.*

For positive numbers  $(\eta_f)_{f \in F}$ , write  $\eta(I) = \prod_{f \in I} \eta_f$  and put

$$\theta_s = \max_{f \in F} \frac{\lambda_f(s)}{\eta_f} \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I). \quad (29)$$

**Theorem 3.3** (Powered matrix-commutative local lemma). *Fix  $s > 0$  and suppose that the powered kernels satisfy (19). Then every deterministic nonanticipating selector  $\Lambda$  satisfies*

$$Z_T^\Lambda(s) \leq \gamma_s a_T(s) \quad (T \geq 0), \quad (30)$$

*or, equivalently,*

$$\sum_{T \geq 0} Z_T^\Lambda(s) z^T \preceq_{\text{coeff}} \frac{\gamma_s}{P_D(z\boldsymbol{\lambda}(s))}. \quad (31)$$

*If  $q\boldsymbol{\lambda}(s)$  lies in the open Shearer region for some  $q > 1$ , then*

$$Z_T^\Lambda(s) \leq \frac{\gamma_s}{P_D(q\boldsymbol{\lambda}(s))} q^{-T}. \quad (32)$$

*If instead  $\theta_s < 1$ , then*

$$Z_T^\Lambda(s) \leq \gamma_s \left( \sum_{R \in \text{Ind}(F)} \eta(R) \right) \theta_s^T \quad (T \geq 0). \quad (33)$$

*All quantities on the right are independent of the selector.*

The parameter-wise trace bound is the sharpest of these statements. The Shearer condition turns it into an exponential estimate, while the cluster expression is a convenient local sufficient condition. At  $s = 1$  they recover established commutative-LLL bounds. The new use is to control the full source-power family, and hence weak randomness and exceptional-tape dimension, by the same matrices.

Whenever either strict condition in Theorem 3.3 holds, choose constants such that

$$Z_T^\Lambda(s) \leq C_s \vartheta_s^T, \quad C_s < \infty, \quad 0 < \vartheta_s < 1, \quad (34)$$

using the corresponding constants from (32) or (33). If the rate supplied by the theorem is zero, the same estimate remains valid after replacing it by any fixed number in  $(0, 1)$ .

**Corollary 3.4** (Dimension and pointwise coding). *Assume (34) and  $\rho_{\max} = \max_{\sigma, f, a} \rho_f(a \mid \sigma) < 1$ . Then*

$$\dim_H^{P_{\text{tape}}} \mathcal{N} \leq s. \quad (35)$$

*Suppose in addition that  $s$ , the process, the selector, and computable numbers  $\widehat{C}_s \geq C_s$  and  $0 < \widehat{\vartheta}_s < 1$  with  $\widehat{\vartheta}_s \geq \vartheta_s$  are computable. Include fixed descriptions of these objects in instance. Every  $w \in \mathcal{L}_T^\Lambda$  then satisfies*

$$K(w \mid T, \text{instance}) \leq s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1), \quad (36)$$

*and every  $\omega \in \mathcal{N}$  has  $\text{Dim}_{\text{eff}, P}(\omega) \leq s$  in this action-outcome source tree.*

The metric in this corollary is tied to complete oracle actions. A binary random-bit implementation gives the same conclusion when it is related to this tree by a computable prefix-free sampler with bounded distortion; no invariance under arbitrary implementations is asserted.

**Corollary 3.5** (Dependent repair choices). *Let  $\nu$  be a possibly history-dependent law that uses only actions available at the corresponding reference history. If a constant  $D < \infty$  satisfies*

$$\nu[w] \leq D P_{\text{tape}}[w]^s \quad (w \in \mathcal{L}_T^\Lambda, T \geq 0), \quad (37)$$

*then*

$$\mathbb{P}_\nu\{\tau \geq T\} \leq D C_s \vartheta_s^T. \quad (38)$$

When  $0 < s \leq 1$  and the reference oracle is uniform on the same  $M_t$  actions available at step  $t$ , the explicit condition

$$\max_a \mathbb{P}_\nu\{A_t = a \mid \mathcal{H}_{t-1}\} \leq M_t^{-s}$$

for every feasible positive-probability history implies (37) with  $D = 1$  when  $\nu$  and  $P_{\text{tape}}$  also use the same initial law. The repair choices may be correlated across time, but may alter neither the selector nor the valid action sets. For a different initial law, or for  $s > 1$ , the corresponding initial likelihood ratio is absorbed into  $D$  when it is bounded.

### 3.1 The variable model and two applications

For a variable-model event  $A$ , let  $v(A)$  be its resampled coordinates and let  $\pi_{v(A)}$  be their product law. Define its local source-power mass by

$$r_A(s) = \sum_{b \in A} \pi_{v(A)}(b)^s. \quad (39)$$

Here  $A$  is identified with its local configurations on  $v(A)$ . Taking  $v_s(\sigma)$  proportional to  $\pi(\sigma)^s$  in (23) gives the exact local charge

$$\lambda_A(s) = r_A(s). \quad (40)$$

Thus Theorem 3.3 can be checked from event scopes, local source masses, and the causality graph, without enumerating all assignments. Replacing the independent-set sum in (29) by  $\prod_{B \in \Gamma(A)} (1 + \eta_B)$  gives a simpler polynomial-time sufficient test, though generally a weaker one. The independent-set refinement is the standard cluster-expansion form (Bissacot et al., 2011; Pegden, 2014); the use made here of the full source-power family is separate.

For a finite graph  $D$ , let

$$\kappa_D = \limsup_{T \rightarrow \infty} |\{h \in \mathcal{M}(D) : |h| = T\}|^{1/T}. \quad (41)$$

Equivalently,  $\kappa_D = R_D^{-1}$ , where  $R_D$  is the first positive zero of  $z \mapsto P_D(z\mathbf{1})$ .

**Theorem 3.6** (Graph-specific min-entropy threshold for  $k$ -SAT). *Let  $\Phi$  be a  $k$ -CNF with clause-overlap graph  $D$ , start from a fixed assignment, and use any deterministic nonanticipating selector. Let  $P$  be the law of independent uniform  $k$ -bit redraws. If a possibly history-dependent redraw law  $\nu$  satisfies, for some  $D_0 < \infty$  and  $s > 0$ ,*

$$\nu[w] \leq D_0 P[w]^s \quad (42)$$

*for every finite live redraw word  $w$ , then, for every  $\xi > \kappa_D$ ,*

$$\mathbb{P}_\nu\{\tau \geq T\} \leq \frac{D_0 2^n}{P_D(\xi^{-1}\mathbf{1})} (\xi 2^{-ks})^T. \quad (43)$$

*In particular, the tail is exponential whenever*

$$s > \frac{\log_2 \kappa_D}{k}. \quad (44)$$

*If  $h > 0$  and each redraw has conditional min-entropy at least  $h$ , then (43) holds with  $D_0 = 1$  and  $ks = h$ . Consequently  $h > \log_2 \kappa_D$  implies exponential termination. For the uniform reference law,*

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 \kappa_D}{k} \right\}. \quad (45)$$

*If the selector is computable, the same bound holds for  $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff},P}(\omega)$ .*

*If  $D$  has maximum degree at most  $d \geq 2$ , then*

$$\kappa_D \leq c_d, \quad c_d = \frac{d^d}{(d-1)^{d-1}}. \quad (46)$$

*Thus  $h > \log_2 c_d$  is a selector-uniform sufficient condition, and the right side of (45) is at most  $\min\{1, \log_2(c_d)/k\}$ .*

The one-step min-entropy hypothesis implies (42) by the chain rule. The converse can fail: a globally dominated source may move randomness between different times. The tree family below makes the global threshold and the dimension bound sharp, but does not by itself prove sharpness of the one-step conditional threshold  $\log_2 c_d$ .

For an infinite run  $\omega$ , let

$$\text{Rec}(\omega) = \{A : A \text{ is repaired infinitely often on } \omega\}. \quad (47)$$

This set is nonempty because there are finitely many clauses.

**Corollary 3.7** (Pointwise localization to recurrent clauses). *For a computable  $k$ -SAT repair process and every infinite run  $\omega$ ,*

$$\text{Dim}_{\text{eff},P}(\omega) \leq \min \left\{ 1, \frac{\log_2 \kappa_{D[\text{Rec}(\omega)]}}{k} \right\}. \quad (48)$$

*For each nonempty  $R \subseteq F$ , the same right side with  $D[R]$  bounds the Hausdorff dimension of the tapes satisfying  $\text{Rec}(\omega) = R$ .*

This is the precise localization statement behind the motivation in Section 1.1: clauses repaired only finitely often affect a finite prefix but not the asymptotic complexity of that tape.

For  $d \geq 2$ , define

$$A_d = 1 + \frac{d^d}{(d-1)^{d-1}}. \quad (49)$$

The graph-specific theorem is sharper but involves  $P_D$ . The following local relaxation has explicit constants and, relative to the worst-case max-degree bound  $c_d$ , loses at most  $\log_2(A_d/c_d) \leq \log_2(5/4)$  bits per repair.

**Theorem 3.8** (An explicit max-degree tail for  $k$ -SAT). *Let  $\Phi$  be a  $k$ -CNF on  $n$  variables with  $m$  clauses, each containing  $k$  distinct variables and overlapping at most  $d \geq 2$  other clauses. Start from an arbitrary assignment and use any deterministic nonanticipating rule to select a violated clause. Suppose that, at every feasible history, the next  $k$ -bit redraw  $B_t$  has worst-case conditional min-entropy at least  $h$ , in the pointwise sense*

$$\max_b \mathbb{P}\{B_t = b \mid \mathcal{H}_{t-1}\} \leq 2^{-h}. \quad (50)$$

If  $\varepsilon = h - \log_2 A_d > 0$ , then

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left( \frac{d}{d-1} \right)^m 2^{-\varepsilon T}. \quad (51)$$

For independent uniform redraws, this becomes

$$\mathbb{P}\{\tau \geq T\} \leq 2^n \left( \frac{d}{d-1} \right)^m \left( \frac{A_d}{2^k} \right)^T. \quad (52)$$

Writing  $P$  for the uniform independent repair-tape law,

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2 A_d}{k} \right\}. \quad (53)$$

If the selector is computable, the same upper bound holds for  $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega)$ . The tail and Hausdorff-dimension conclusions are uniform over every deterministic nonanticipating selector and every history-dependent redraw law satisfying (50).

**Corollary 3.9** (Fresh fair bits). *Before repair  $t$ , let the past determine an injective map*

$$E_t : \{0, 1\}^r \longrightarrow \{0, 1\}^k, \quad 1 \leq r \leq k,$$

*then read  $r$  fresh independent fair bits  $U_t$  and redraw the selected clause to  $E_t(U_t)$ . If  $\varepsilon_r = r - \log_2 A_d > 0$ , then (51) holds with  $\varepsilon = \varepsilon_r$ , and*

$$\mathbb{E}\tau \leq \left\lceil \frac{n + m \log_2(d/(d-1))}{\varepsilon_r} \right\rceil + \frac{1}{2^{\varepsilon_r} - 1}. \quad (54)$$

*The expected number of fair bits is  $r\mathbb{E}\tau$ . In particular, if  $A_d \leq 2^{k-1}$ , taking  $r = \lceil \log_2 A_d + 1 \rceil$  gives*

$$\mathbb{E}[\text{fair bits}] = O((n + m/d) \log(d+1)). \quad (55)$$

*The redraw map may change adaptively with the entire past, but must be fixed before  $U_t$  is read.*

**Corollary 3.10** (A Santha–Vazirani tape). *Let the bits  $(U_j)$  satisfy, for some  $0 < \alpha \leq 1/2$ ,*

$$\alpha \leq \mathbb{P}\{U_j = 1 \mid U_1, \dots, U_{j-1}\} \leq 1 - \alpha$$

*at every positive-probability history (Santha and Vazirani, 1986). Feed successive blocks of  $r$  bits through the adaptive injective maps in Corollary 3.9. If*

$$\varepsilon_\alpha = r \log_2 \frac{1}{1 - \alpha} - \log_2 A_d > 0,$$

*then (51) and (54) hold with  $\varepsilon_\alpha$  in place of  $\varepsilon_r$ . Taking  $r = k$  and  $E_t$  to be the identity runs ordinary clause resampling directly from the Santha–Vazirani tape, without an extractor.*

**Proposition 3.11** (A one-step min-entropy obstruction). *Let  $d \geq 2$  and  $k \geq 2$ . If  $d + 1 \leq 2^{k-1}$ , there is a satisfiable  $k$ -CNF of dependency degree  $d$  whose repair scopes overlap without being equal. From a specified initial assignment, uniform independent  $k$ -bit redraws satisfy*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \frac{d + 1}{2^k}, \quad (56)$$

$$\dim_H^P \mathcal{N} = \sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \frac{\log_2(d + 1)}{k}. \quad (57)$$

*There is also a fixed clause-wise redraw source of conditional block min-entropy  $\log_2(d + 1)$  on which every run is infinite. The dependency graph of this instance is  $K_{d+1}$ , for which  $\kappa_{K_{d+1}} = d + 1$ . Thus Theorem 3.6 is sharp for this graph: strict inequality above  $\log_2(d + 1)$  gives exponential termination, whereas equality can sustain an infinite run. Among guarantees that use only the maximum degree  $d$ , the one-step sufficient and obstructing thresholds are separated by less than*

$$\log_2 c_d - \log_2(d + 1) < \log_2 e$$

*bits. When  $d + 1 = 2^r$ , the infinite-run source is an injective  $r$ -fair-bit redraw. This last source need not be Santha–Vazirani, so the proposition does not claim sharpness for Corollary 3.10.*

**Theorem 3.12** (Tree formulas approach the max-degree boundary). *Fix integers  $2 \leq d \leq k$ . For every height  $\ell \geq 1$ , there is a satisfiable  $k$ -CNF  $\Phi_{d,k,\ell}$  whose clause-overlap graph is the full rooted  $(d - 1)$ -ary tree  $T_{d,\ell}$  of height  $\ell$ . Every variable occurs in at most two clauses, every clause overlaps at most  $d$  others, and every pair of adjacent bad events is disjoint. Draw the initial assignment uniformly, and then use independent uniform redraws; let  $P$  denote this sequential tape law. For any deterministic current-assignment selector,*

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = 2^{-k} \kappa_{T_{d,\ell}}, \quad (58)$$

$$\dim_H^P \mathcal{N} = \frac{\log_2 \kappa_{T_{d,\ell}}}{k}. \quad (59)$$

*If the selector is computable, the right side of (59) is also the maximal effective strong dimension. Moreover,*

$$\kappa_{T_{d,\ell}} \uparrow \frac{d^d}{(d - 1)^{d-1}} = c_d \quad (\ell \rightarrow \infty). \quad (60)$$

*Thus the max-degree dimension bound in Theorem 3.6 is best possible in the supremum over finite instances, already for tree dependency graphs.*

For each fixed  $\ell$ , put  $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$ . There are a probability law  $\nu_{\delta_\ell}$  supported on  $\mathcal{N}$  and a finite  $D_\ell$  such that

$$\nu_{\delta_\ell}[w] \leq D_\ell P[w]^{\delta_\ell} \quad (61)$$

for every finite tape prefix  $w$ . Consequently the same law satisfies  $\nu_{\delta_\ell}[w] \leq D_\ell P[w]^s$  for every  $0 \leq s \leq \delta_\ell$ . Hence the global cylinder threshold (44) is sharp, including at the critical power. This statement does not assert a matching one-step conditional min-entropy law.

The clique and tree lower families are extremal. The clique has partially overlapping scopes and gives the exact one-step threshold for its graph; the trees make the maximum-degree dimension and global source threshold sharp. For general degree- $d$  repairs with  $d+1 \leq 2^{k-1}$ , the present one-step lower and upper thresholds are  $\log_2(d+1)$  and  $\log_2 c_d$  bits. Because both lower families are extremal, a matching lower bound for nonextremal overlapping repairs remains open.

These are robustness results for the original local repair map, not stronger satisfiability criteria. The trace form gives the sharp graph-specific upper bound, while Theorem 3.8 trades at most  $\log_2(5/4)$  additional bits relative to the worst-case max-degree bound for an explicit prefactor and the end-to-end randomness estimate. Proposition 3.11 also shows why logarithmic fresh-block randomness cannot be removed in that model. Randomness reuse and derandomization can have different guarantees. Viewing the same trace bound across source powers also yields the dimension and pointwise-localization statements. For uniform fixed-arity redraws this is equivalent to rescaling the  $s = 1$  history count; the powered formulation also handles nonuniform and state-dependent repair laws.

**Corollary 3.13** (Rainbow perfect matchings). *Let  $n \geq 2$ . Color the edges of  $K_{2n}$  so that each color occurs on at most  $q \geq 2$  edges, and run Kolmogorov's perfect-matching repair oracle from an arbitrary perfect matching. A flaw is a pair of vertex-disjoint edges of the same color. Put*

$$A = (2n-1)(2n-3), \quad C = (2n-1)(q-1).$$

*Suppose, for some  $0 \leq h \leq \log_2 A$ , every repair choice has conditional atom probability at most  $2^{-h}$  at every feasible positive-probability history. If, for some  $\varepsilon > 0$ ,*

$$h \geq \log_2 \left( \frac{256}{27} C \right) + \varepsilon,$$

*then every deterministic nonanticipating selector satisfies*

$$\mathbb{P}\{\tau \geq T\} \leq (2n-1)!! e^{n/6} 2^{-\varepsilon T}. \quad (62)$$

*For uniform repair choices this is nonvacuous when  $q-1 < 27(2n-3)/256$ . Writing  $P$  for the uniform oracle-action law,*

$$\dim_H^P \mathcal{N} \leq \min \left\{ 1, \frac{\log_2((256/27)(2n-1)(q-1))}{\log_2((2n-1)(2n-3))} \right\}. \quad (63)$$

*The same bound holds for  $\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff},P}(\omega)$  under the computability assumptions of Corollary 3.4.*

The matching oracle is a non-product-space example: its action count, atomicity, and strong commutativity are established by Kolmogorov (2018). The dimensions in (63) refer to its uniform action-outcome tree. The derivations of both applications are given in Appendix B.

*Remark 3.14* (Expected repairs). As an elementary consequence, any tail bound  $\mathbb{P}\{\tau \geq T\} \leq C_0 2^{-\varepsilon T}$  with  $C_0 \geq 1$  gives

$$\mathbb{E}\tau \leq \left\lceil \frac{\log_2 C_0}{\varepsilon} \right\rceil + \frac{1}{2^\varepsilon - 1}. \quad (64)$$

Indeed, bound the first  $\lceil \log_2(C_0)/\varepsilon \rceil$  terms in  $\mathbb{E}\tau = \sum_{T \geq 1} \mathbb{P}\{\tau \geq T\}$  by one and sum the remaining geometric series. For  $0 < \varepsilon \leq 1$ , the  $k$ -SAT bound therefore gives  $\mathbb{E}\tau = O((n + m/d)/\varepsilon)$ , and the rainbow-matching bound gives  $\mathbb{E}\tau = O(n \log n/\varepsilon)$ .

*Remark 3.15* (Ordinary and powered commutativity). Matrix commutativity at  $s = 1$  does not imply commutativity after entrywise powering. For example,

$$A_0 = \begin{pmatrix} 4/5 & 1/5 \\ 2/5 & 3/5 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 7/10 & 3/10 \\ 3/5 & 2/5 \end{pmatrix}$$

satisfy  $A_0 B_0 = B_0 A_0$ , whereas their entrywise square roots do not commute. Thus ordinary matrix commutation need not control the full power family. Product-preserving action diamonds do control it, and so does ordinary matrix commutation when every enabled action of a fixed flaw is uniform and the number of actions does not depend on the state.

## 4 Exact source-power calculations

The central theorem gives selector-uniform upper bounds. This section turns to processes for which the survival law and both dimensions can be computed exactly. Rejection sampling fixes the normalization in one state; a general finite-state reduction then isolates the information that an exact smaller state must preserve. Two local-repair families show how this reduction can be realized by trace growth or by a low-dimensional statistic.

**One-state calibration.** Consider the simplest deterministic form of rejection sampling. Draw  $X_1, X_2, \dots$  independently from a full-support law  $\pi$  on a finite set  $\Omega$ , reject a draw in  $U \subseteq \Omega$ , and return the first draw in  $G = \Omega \setminus U$ . This is a special case of the classical accept–reject method ([von Neumann, 1951](#); [Devroye, 1986](#)). Write  $p = \pi(U)$ . If  $0 < p < 1$  and  $\tau$  is the number of rejected draws, then

$$\mathbb{P}\{\tau \geq n\} = p^n, \quad \mathbb{P}\{\tau = t\} = p^t(1 - p), \quad \mathbb{E}\tau = \frac{p}{1 - p}. \quad (65)$$

The returned value has law  $\pi(\cdot \mid G)$ . The algorithm terminates almost surely, but its nonterminating set is the nonempty null set  $\mathcal{N} = U^\mathbb{N}$ .

This one-state process calibrates the two notions of tape dimension. In general, effective strong dimension is associated with packing rather than Hausdorff dimension ([Athreya et al., 2007](#)); their equality here is a property of a regular product set, not a general identity. If  $U$  is nonempty, let  $\delta \in [0, 1]$  be the unique solution of

$$\sum_{x \in U} \pi(x)^\delta = 1. \quad (66)$$

Then

$$\sup_{\omega \in \mathcal{N}} \text{Dim}_{\text{eff}, P}(\omega) = \dim_H^P \mathcal{N} = \delta, \quad \text{Dim}_{\text{eff}, P}(\omega) \leq \delta \quad (\omega \in \mathcal{N}). \quad (67)$$

Under computability the supremum is attained. Indeed, writing  $\mathbf{1}\{\cdot\}$  for an indicator, the tilted law

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}$$

is a computable probability law, and every tape that is Martin–Löf random under  $Q_\delta^\mathbb{N}$  attains  $\delta$  by the Levin–Schnorr theorem (Li and Vitányi, 2019). Under a uniform  $q$ -ary source,  $\pi(U) = q^{\delta-1}$ , so the tail exponent and dimension determine one another; for a nonuniform source they need not.

For example, take  $\Omega = \{00, 01, 10, 11\}$  uniformly and  $U = \{00, 01\}$ . Every surviving block has its first bit fixed and its second bit free, so both the maximal effective strong dimension and the Hausdorff dimension are  $1/2$ . Proposition A.2 in Appendix A gives the corresponding Rényi-entropy identity for a nonuniform source. These facts use standard product geometry and are not claimed as a new rejection-sampling theorem; their role is to fix, in one state, the survival and dimension normalizations used below.

#### 4.1 Finite-state reduction

Let  $\mathcal{X}$  be a finite set of live states. From  $x \in \mathcal{X}$ , a source outcome  $e$  of probability  $p(e) > 0$  either moves the computation to a live state  $F(x, e)$  or terminates it. For  $s \geq 0$ , let

$$(B_s)_{xy} = \sum_{e:F(x,e)=y} p(e)^s, \quad x, y \in \mathcal{X}. \quad (68)$$

Thus  $B_1$  is the usual killed transition matrix, whereas the family  $(B_s)_{s \geq 0}$  counts source-weighted tape prefixes.

The following standard finite-state pressure formula is stated here because it is the bridge from the operators to the three quantities studied in the paper. Its detailed form and proof appear in Appendix A.

**Proposition 4.1** (Finite-state pressure). *Fix an initial live state. Suppose its reachable live-state graph is finite, strongly connected, contains a directed cycle, and satisfies  $\max_e p(e) < 1$ . There is a unique  $\delta \in [0, 1]$  such that  $\rho(B_\delta) = 1$ . The survival exponent is  $-\log_2 \rho(B_1)$ , and the source-relative Hausdorff dimension of the nonterminating tapes is  $\delta$ . If the system and its critical Perron law are computable, every nonterminating tape has effective strong dimension at most  $\delta$ , and some tape attains  $\delta$ . For a reducible graph, these statements apply componentwise to reachable cyclic strongly connected components.*

Let  $\phi : \mathcal{X} \rightarrow \mathcal{C}$  be a surjection. Say that  $\phi$  *lumps the source powers* if, for every  $s \geq 0$  and  $c, d \in \mathcal{C}$ , the quantity

$$R_s(c, d) := \sum_{e:F(x,e) \in \phi^{-1}(d)} p(e)^s \quad (69)$$

does not depend on the choice of  $x \in \phi^{-1}(c)$ . This phrase is descriptive rather than standard. At  $s = 1$ , condition (69) is the usual strong lumpability condition for a Markov chain (Kemeny and Snell, 1960); requiring it for all  $s$  is what also preserves the geometry of the source tape.

**Proposition 4.2** (Exact source-power reduction). *If  $\phi$  satisfies (69), then, for every  $x \in \mathcal{X}$  and integer  $T \geq 0$ ,*

$$(B_s^T \mathbf{1})(x) = (R_s^T \mathbf{1})(\phi(x)). \quad (70)$$

*In particular,*

$$\rho(B_s) = \rho(R_s). \quad (71)$$

*If  $R_s r_s = \varrho_s r_s$  for a strictly positive  $r_s$  and  $\varrho_s > 0$ , then  $h_s = r_s \circ \phi$  satisfies  $B_s h_s = \varrho_s h_s$ , and*

$$Q_s(e \mid x) = \frac{p(e)^s h_s(F(x, e))}{\varrho_s h_s(x)} \quad (72)$$

for outcomes with  $F(x, e) \in \mathcal{X}$  (and zero otherwise) is a probability law on live transitions. Every path  $w = e_1 \cdots e_T$  from  $x_0$  to  $x_T$  then obeys

$$\log_2 \frac{Q_s[w]}{P[w]} = (1 - s) \log_2 \frac{1}{P[w]} - T \log_2 \varrho_s + \log_2 \frac{h_s(x_T)}{h_s(x_0)}. \quad (73)$$

Consequently, under the hypotheses of Proposition 4.1, the survival exponent is  $-\log_2 \rho(R_1)$  and the nontermination dimension is the solution of  $\rho(R_s) = 1$ . The same reduced eigenfunction gives the dimension-attaining Doob transform and the effective-dimension bound.

The proof is given in Appendix D.

The continuum of powers in (69) is not an algorithmic obstacle when the transition system is explicit. Let  $\mathcal{R}$  be the finite set of values taken by  $p(e)$ . Distinct functions  $r^s$ ,  $r \in \mathcal{R}$ , are linearly independent as functions of  $s$ . Hence (69) is equivalent to requiring, for each  $r \in \mathcal{R}$ , that the number of probability- $r$  outcomes from  $x$  into each class  $d$  depend only on  $\phi(x)$ . When the probabilities are given by exact labels with decidable equality—for example, as rational or algebraic numbers—ordinary labelled partition refinement therefore finds the coarsest source-power lumping in time polynomial in the *explicit* transition graph. This does not by itself avoid an exponentially large assignment graph; it identifies exactly what a smaller representation must preserve.

There is a simple local criterion for Proposition 4.2. Suppose an assignment has an additive statistic  $g(x) = \sum_v g_v(x_v)$  and that terminal status depends only on  $g(x)$ . If, whenever  $g(x) = k$ , the selected repair block has a fixed old score  $c(k)$  and the total  $s$ -weight of its redraws with new score  $j$  is a number  $\nu_{s,k}(j)$  independent of  $x$ , then  $g$  lumps the source powers, with

$$R_s(k, k - c(k) + j) = \nu_{s,k}(j). \quad (74)$$

This criterion allows the selected block itself to depend on the full assignment. Only its contribution to the statistic, and the source-weighted law of its replacement, must be determined by  $k$ .

Two mechanisms now give exact source-power sums for local repair. Extremal repair is enumerated by commuting-event traces; disagreement repair admits the all-power reduction of Proposition 4.2. The second case also separates the state needed for survival and dimension from the state needed to continue the computation exactly.

## 4.2 Extremal repair and the Shearer boundary

The next result computes survival and both tape dimensions for every finite extremal instance of partial rejection sampling. Let  $(X_v)_{v \in V}$  be independent finite-valued variables with full-support laws  $\pi_v$ . There is at least one bad event, and  $A_i$  depends on the scope  $V_i \subseteq V$  and is represented by a set  $\mathcal{B}_i \subseteq \prod_{v \in V_i} \mathcal{X}_v$  of bad local configurations. The dependency graph  $D$  has vertex set  $[m]$  and joins  $i$  and  $j$  when  $V_i \cap V_j \neq \emptyset$ . The instance is *extremal* if

$$A_i \cap A_j = \emptyset \quad (ij \in E(D)).$$

Thus two events are either supported on disjoint variables or cannot occur simultaneously. Assume that the satisfying set  $\Sigma = \bigcap_i A_i^c$  is nonempty and that every  $A_i$  is nonempty. At each step, a fixed deterministic current-assignment rule selects a currently true event and redraws the variables in its scope. Let  $P$  be the resulting sequential source law. Assume also that every bad local configuration has source probability strictly between zero and one. Only the values actually drawn, in their order of revelation, form the tape. This is the consumed-choice tree of Section 2.1; unused entries of a resampling table are not additional tape coordinates.

For  $s \geq 0$ , define the source-power mass of event  $i$  and of the satisfying frontier by

$$r_i(s) = \sum_{b \in \mathcal{B}_i} \pi_{V_i}(b)^s, \quad Z_\Sigma(s) = \sum_{x \in \Sigma} \pi(x)^s. \quad (75)$$

In particular,  $r_i(1) = p_i := \mathbb{P}(A_i)$ . If  $A_i$  is atomic, meaning that  $\mathcal{B}_i$  is a singleton, then  $r_i(s) = p_i^s$ .

Recall the signed independence polynomial  $P_D$  and the open Shearer region from (26); the boundary is where its downward positivity first fails along a positive ray (Shearer, 1985; Scott and Sokal, 2005). In the trace monoid  $\mathcal{M}(D)$ , letters  $i$  and  $j$  commute exactly when  $ij \notin E(D)$ . Write  $N_i(h)$  for the number of occurrences of  $i$  in a trace  $h$ , and  $|h| = \sum_i N_i(h)$ . Set

$$a_T(s) = \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_{i=1}^m r_i(s)^{N_i(h)}. \quad (76)$$

**Theorem 4.3** (Extremal source-power formula). *Let  $\mathcal{H}_T$  be the set of minimal consumed tape prefixes ending when the algorithm halts after exactly  $T$  repairs. Then, as an identity of formal power series,*

$$\sum_{w \in \mathcal{H}_T} P[w]^s = Z_\Sigma(s) a_T(s), \quad (77)$$

$$\sum_{T \geq 0} u^T \sum_{w \in \mathcal{H}_T} P[w]^s = \frac{Z_\Sigma(s)}{P_D(ur_1(s), \dots, ur_m(s))}. \quad (78)$$

Let  $\mathcal{L}_T$  be the set of prefixes for which the computation is still live after  $T$  repairs and put  $L_T(s) = \sum_{w \in \mathcal{L}_T} P[w]^s$ . For every fixed  $s \geq 0$  there are constants  $0 < c_s \leq C_s < \infty$ , independent of  $T$ , such that

$$c_s a_T(s) \leq L_T(s) \leq C_s a_T(s), \quad T \geq 1. \quad (79)$$

Here  $L_T(1) = \mathbb{P}\{\tau > T\}$ . Replacing  $T$  by  $T - 1$  gives the convention  $\mathbb{P}\{\tau \geq T\}$  used for the survival exponent; this shift changes neither an exponential rate nor a dimension.

Let  $R(s)$  be the radius of convergence in  $u$  of the right-hand side of (78). Equivalently,  $R(s)$  is the first positive zero of  $u \mapsto P_D(u\mathbf{r}(s))$ . There is a unique  $\delta \in [0, 1]$  such that  $R(\delta) = 1$ , and

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = R(1)^{-1}, \quad \dim_H^P \mathcal{N} = \delta. \quad (80)$$

Moreover, as a formal power-series identity, and analytically for  $|u| < R(1)$ ,

$$\mathbb{E}u^\tau = \frac{P_D(\mathbf{p})}{P_D(u\mathbf{p})}. \quad (81)$$

The vector  $\mathbf{r}(\delta)$  lies on the boundary of Shearer's region for  $D$ ; more explicitly,

$$P_D(\mathbf{r}(\delta)) = 0, \quad P_D(u\mathbf{r}(\delta)) > 0 \quad (0 \leq u < 1). \quad (82)$$

If the source, events, and selection rule are computable, the largest effective strong dimension of a nonterminating tape is also  $\delta$ , and this value is attained.

The proof is given in Appendix D.

The exact source-power masses also give a simple comparison using coarser event data. For a positive vector  $\mathbf{z}$ , let  $\mathcal{R}_D(\mathbf{z})$  denote the first positive zero of  $u \mapsto P_D(u\mathbf{z})$ .

**Corollary 4.4** (Marginal–multiplicity bounds). Put  $k_i = |\mathcal{B}_i|$ . Let  $\delta_-$  and  $\delta_+$  be the unique solutions in  $[0, 1]$  of

$$\mathcal{R}_D((p_i^{\delta_-})_i) = 1, \quad \mathcal{R}_D((k_i^{1-\delta_+} p_i^{\delta_+})_i) = 1. \quad (83)$$

Then

$$\delta_- \leq \dim_H^P \mathcal{N} \leq \delta_+. \quad (84)$$

Let

$$\kappa_D = \limsup_{T \rightarrow \infty} |\{h \in \mathcal{M}(D) : |h| = T\}|^{1/T}$$

be the unweighted trace-growth rate. If all  $p_i = p$  and all  $k_i = k$ , this becomes

$$\frac{\log_2 \kappa_D}{\log_2(1/p)} \leq \dim_H^P \mathcal{N} \leq \frac{\log_2(k \kappa_D)}{\log_2(k/p)}. \quad (85)$$

Both bounds coincide when the events are atomic.

The proof is given in Appendix D.

For atomic events, Theorem 4.3 takes a particularly simple form.

**Corollary 4.5** (Powered-marginal Shearer boundary). If every bad event is atomic, then

$$\dim_H^P \mathcal{N} = \delta, \quad \mathbf{p}^\delta = (p_1^\delta, \dots, p_m^\delta) \text{ lies on the Shearer boundary of } D. \quad (86)$$

The same  $\delta$  is the largest effective strong dimension under computability. If, in addition,  $p_i = p$  for every  $i$ , put

$$\alpha = - \lim_{T \rightarrow \infty} \frac{1}{T} \log_2 \mathbb{P}\{\tau \geq T\}.$$

Then

$$\mathbb{P}\{\tau \geq T\}^{1/T} \rightarrow p \kappa_D, \quad \delta = \frac{\log_2 \kappa_D}{\log_2(1/p)} = 1 - \frac{\alpha}{\log_2(1/p)}. \quad (87)$$

For unequal atomic probabilities, with  $\ell_{\min} = \min_i \log_2(1/p_i)$  and  $\ell_{\max} = \max_i \log_2(1/p_i)$ ,

$$1 - \frac{\alpha}{\ell_{\min}} \leq \delta \leq 1 - \frac{\alpha}{\ell_{\max}}. \quad (88)$$

The proof is given in Appendix D.

### 4.3 Computing the critical power

The exact formula does not make the critical power easy to compute from an arbitrary local instance. Consider the following promised problem, POWERED-EXTREMAL-DIMENSION. The input is an explicitly listed atomic extremal binary variable-model instance, a positive integer  $L$  in unary, and rationals  $z_i \in (0, 1)$ , under the promise that the marginal of event  $i$  is  $p_i = z_i^L$ . The question is whether  $\dim_H^P \mathcal{N} < 1/L$ .

**Theorem 4.6** (Hardness of the exact critical power). POWERED-EXTREMAL-DIMENSION is  $\#P$ -hard under polynomial-time Turing reductions, even when the selector always repairs the lowest-index present event and

$$p_i \leq 2^{-(d_i+2)}, \quad (89)$$

where  $d_i$  is the event's dependency degree. In the hard family, every source variable is binary with a rational law, every event is atomic, and all but one event have scope at most six; the remaining event may have linear scope. The same hardness holds with Hausdorff dimension replaced by maximal effective strong dimension.

The proof is in Appendix D. It is an exact-comparison result, not a constant-additive approximation result, and it does not establish hardness when every event has bounded scope.

The next theorem supplies the missing bounded-scope boundary. It is a qualitative classification rather than a hardness-of-approximation result.

**Proposition 4.7** (The qualitative boundary in the finite variable model). *Consider any finite full-support product source on  $n$  variables and any deterministic nonanticipating rule for selecting a present bad event and redrawing its scope. A flawless assignment exists if and only if the algorithm has an exponential termination tail from every initial assignment. More precisely, for a flawless assignment  $x^*$ , put*

$$\beta = \min_A \pi_{v(A)}(x_{v(A)}^*) > 0.$$

Then

$$\mathbb{P}\{\tau > qn\} \leq (1 - \beta^n)^q \quad (q \in \mathbb{Z}_{\geq 0}). \quad (90)$$

If no flawless assignment exists, every tape is nonterminating.

**Theorem 4.8** (Local hardness at the termination boundary). *Let  $\Phi$  be an exact 3-CNF formula with three distinct, non-tautological literals in each clause. Start from a full-support initial law on assignments and always resample the three variables of the lowest-index violated clause with fresh fair bits. Let  $P_\Phi$  be the substochastic transition matrix on flawed assignments, put  $\lambda_\Phi = \rho(P_\Phi)$ , and let  $\delta_\Phi$  be the source-relative Hausdorff dimension of its nonterminating consumed tapes. Then*

$$\delta_\Phi = 1 + \frac{1}{3} \log_2 \lambda_\Phi \quad (91)$$

whenever the live state space is nonempty, and

$$\Phi \in \text{SAT} \iff \lambda_\Phi < 1 \iff \delta_\Phi < 1. \quad (92)$$

The latter two decision problems are NP-complete even when every variable occurs in at most four clauses. Under this restriction, every repair has scope three and the dependency graph has maximum degree at most nine. The equality tests  $\lambda_\Phi = 1$  and  $\delta_\Phi = 1$  are coNP-complete. The succinct input to each decision problem is the ordered formula  $\Phi$ ; the exponentially large matrix  $P_\Phi$  is defined by  $\Phi$  and is not part of the input.

The proofs are in Appendix D. Thus the strict boundary in (92) expresses ordinary CSP feasibility. Its value is to show that even the coarsest exact pressure question is hard under genuinely local restrictions; it is not a constant-gap lower bound. The quantitative value of  $\lambda_\Phi$  or  $\delta_\Phi$  below one is the more informative computational problem.

**Relation to classical trace enumeration.** The confluence of extremal partial rejection sampling, the correspondence between transcripts and Cartier–Foata traces, and the identity  $1/P_D$  are classical (Cartier and Foata, 1969; Guo et al., 2019; Jerrum, 2024). Jerrum’s exact probability-weighted transcript series gives Equation (81) directly (Jerrum, 2024). Moreover, Abbes and Mairesse (2015) construct Bernoulli measures on irreducible infinite trace boundaries from Möbius vanishing and positivity. The additional step here is to power every consumed source value before summing and to treat reducible graphs componentwise. This identifies the source-power pressure, and hence the Hausdorff and pointwise dimensions of nontermination, with the point at which  $\mathbf{r}(s)$  reaches the same boundary mechanism. For non-atomic events it is important to use  $r_i(s)$  from (75); replacing it by  $p_i^s$  is generally incorrect (indeed  $r_i(s) \geq p_i^s$  for  $0 \leq s \leq 1$ ).

#### 4.4 A nonextremal exact reduction

Let  $G = (V, E)$  be any connected simple graph with  $N = |V| \geq 2$ . Draw independent  $X_v \sim \text{Bernoulli}(p)$ , where  $0 < p < 1$ , and put  $q = 1 - p$ . For each edge  $e = \{u, v\}$ , define

$$A_e = \{X_u \neq X_v\}.$$

Fix any deterministic nonanticipating rule that selects a true event; the rule may use the current assignment and the past repair history, but not future source outcomes. To repair  $A_e$ , redraw  $X_u$  and  $X_v$  independently from their source laws. The terminal assignments are  $0^V$  and  $1^V$ . When  $G$  has two adjacent edges, the repairs genuinely overlap. For two distinct edges sharing one endpoint,

$$\mathbb{P}(A_e) = 2pq, \quad \mathbb{P}(A_e \cap A_f) = pq. \quad (93)$$

Let  $\tau$  be the number of repairs before a constant assignment is reached, with  $\tau = 0$  if the initial assignment is constant. Let  $P$  be the sequential source law of the initial assignment and all redraws, let  $\mathcal{N}(G, p) = \{\tau = \infty\}$ , and for a redraw word  $w = z_1 \cdots z_T$  write  $P_p[w] = \prod_{t=1}^T P_p(z_t)$ , where  $P_p(z) = p^{z_1+z_2} q^{2-z_1-z_2}$  for  $z = (z_1, z_2)$ . For  $s \geq 0$ , define the  $(N-1) \times (N-1)$  tridiagonal matrix  $R_s$  on  $\{1, \dots, N-1\}$  by

$$R_s(k, k-1) = q^{2s}, \quad R_s(k, k) = 2(pq)^s, \quad R_s(k, k+1) = p^{2s}, \quad (94)$$

where entries outside  $\{1, \dots, N-1\}$  are omitted.

**Theorem 4.9** (Exact nontermination for disagreement repair). *For every connected  $G$ , every deterministic nonanticipating selection rule, and every integer  $T \geq 0$ , the entire law of  $\tau$  is the same. More precisely,*

$$\mathbb{P}\{\tau > T\} = \sum_{k=1}^{N-1} \binom{N}{k} p^k q^{N-k} (R_1^T \mathbf{1})_k. \quad (95)$$

More generally, the source-power sum of all tapes whose initial assignment and next  $T$  redraws remain live is

$$Z_T(s) = \sum_{k=1}^{N-1} \binom{N}{k} p^{sk} q^{s(N-k)} (R_s^T \mathbf{1})_k. \quad (96)$$

Thus  $Z_T(1) = \mathbb{P}\{\tau > T\}$ . Replacing  $T$  by  $T-1$  recovers the convention  $\mathbb{P}\{\tau \geq T\}$ ; this shift does not affect any exponential rate or dimension.

Set

$$\vartheta = \frac{\pi}{N}, \quad c_N = \cos\left(\frac{\pi}{2N}\right).$$

The exact source-power rate, survival rate, and source-relative dimension are

$$\varrho_s = \lim_{T \rightarrow \infty} Z_T(s)^{1/T} = \rho(R_s) = 4(pq)^s c_N^2, \quad (97)$$

$$\lambda_{N,p} = \lim_{T \rightarrow \infty} \mathbb{P}\{\tau > T\}^{1/T} = 4pq c_N^2, \quad (98)$$

$$\dim_H^P \mathcal{N}(G, p) = \delta_{N,p} = \frac{\log(4c_N^2)}{\log(1/(pq))}. \quad (99)$$

For computable data, including computable  $p$  and a computable selection rule,  $\delta_{N,p}$  is also the largest effective strong dimension of a nonterminating tape, and it is attained.

The dimension-attaining Doob transform is explicit. Start from any fixed computable live assignment; this finite initial choice does not affect dimension. If the current assignment has  $k$  ones and a redraw  $z \in \{00, 01, 10, 11\}$  has  $j(z)$  ones, put  $d(z) = j(z) - 1$  and

$$Q^*(z | k) = \frac{\sin((k + d(z))\vartheta)}{4c_N^2 \sin(k\vartheta)}, \quad (100)$$

with value zero when  $k + d(z) \in \{0, N\}$ . This conditional law for the four redraw outcomes is independent of  $p$ , the graph, and the selection rule. Its Hamming-weight marginal is reversible with stationary distribution

$$\nu^*(k) = \frac{2}{N} \sin^2(k\vartheta), \quad 1 \leq k < N. \quad (101)$$

For every  $s \geq 0$  and every surviving path  $w = z_1 \cdots z_T$  from  $k_0$  to  $k_T$ ,

$$\log_2 \frac{Q^*[w | k_0]}{P_p[w]} = (1 - s)I_p(w) - T \log_2 \varrho_s + \log_2 \frac{h_s(k_T)}{h_s(k_0)}, \quad (102)$$

where

$$h_s(k) = \left(\frac{q}{p}\right)^{sk} \sin(k\vartheta), \quad I_p(w) = \log_2 \frac{1}{P_p[w]}. \quad (103)$$

Set  $h_s(0) = h_s(N) = 0$  when a boundary value occurs in a recurrence. Thus the same change of measure gives the survival exponent at  $s = 1$  and the dimension bound at  $s = \delta_{N,p}$ .

The proof is given in Appendix D.

There is a particularly simple information identity behind the theorem. For one redraw,

$$P_p(z) = pq \left(\frac{p}{q}\right)^{d(z)}.$$

Consequently every surviving path satisfies

$$I_p(w) = T \log_2 \frac{1}{pq} + (k_T - k_0) \log_2 \frac{q}{p}. \quad (104)$$

The second term is bounded independently of  $T$ . If  $\alpha_{N,p} = -\log_2 \lambda_{N,p}$ , then

$$\delta_{N,p} = 1 - \frac{\alpha_{N,p}}{\log_2(1/(pq))}. \quad (105)$$

Equivalently, if  $c = \log_2(1/(pq))$  is the source information per redraw up to the bounded endpoint term and  $h = \log_2(4c_N^2)$  is the surviving information rate, then  $\alpha_{N,p} = c - h$  and  $\delta_{N,p} = h/c$ . Thus  $\alpha_{N,p} = c(1 - \delta_{N,p})$ .

For a fair source,

$$\lambda_{N,1/2} = c_N^2, \quad \delta_{N,1/2} = 1 + \log_2 c_N, \quad \delta_{N,1/2} = 1 - \frac{\alpha_{N,1/2}}{2}. \quad (106)$$

Moreover,

$$\alpha_{N,1/2} = \frac{\pi^2}{4N^2 \ln 2} + O(N^{-4}), \quad 1 - \delta_{N,1/2} = \frac{\pi^2}{8N^2 \ln 2} + O(N^{-4}). \quad (107)$$

Still under the fair-source assumption  $p = 1/2$ , the mean absorption time is also exact:

$$\mathbb{E}[\tau | M_0 = k] = 2k(N - k), \quad \mathbb{E}\tau = \frac{N(N - 1)}{2}. \quad (108)$$

Thus the process terminates almost surely and has quadratic mean running time, while the dimension of its nonterminating tapes tends to one.

The same result has a standard constraint-satisfaction form. Let  $G = (L \cup R, E)$  be connected and bipartite, draw  $X_v \sim \text{Bernoulli}(p)$  on  $L$  and  $X_v \sim \text{Bernoulli}(q)$  on  $R$ , and declare a monochromatic edge bad. Repairing a bad edge redraws its endpoints from their respective source laws. The change of variables

$$Y_v = X_v \quad (v \in L), \quad Y_v = 1 - X_v \quad (v \in R)$$

turns this process into disagreement repair with i.i.d.  $\text{Bernoulli}(p)$  variables. Therefore Theorem 4.9 gives the exact time and tape geometry of local resampling for a proper two-colouring, for every connected bipartite graph and every deterministic repair rule.

**Counting state versus continuation state.** The statistic  $M$  determines all the quantities in Theorem 4.9, but it does not determine the next edge selected by a specified rule. This distinction cannot be removed merely by assuming that the event–variable incidence graph has small pathwidth.

Two live assignments are *continuation-equivalent* if every common future redraw sequence produces the same repair labels and the same halting decision. Any self-contained deterministic state description must refine this equivalence. Conversely, its equivalence classes form the minimal deterministic continuation machine, by the Myhill–Nerode principle.

**Proposition 4.10** (A pathwidth-one state-cardinality obstruction). *Let  $G$  be the path on vertices  $0, 1, \dots, N-1$ , and always repair the leftmost disagreeing edge. Suppose a self-contained deterministic state summary has two properties: its current value determines the next repair (or halt), and it updates exactly from that value and the realized redraw. Then it has at least  $2^{N-1}$  states when the halting state is included, and at least  $2^{N-1} - 1$  live states.*

The proof is given in Appendix D.

The incidence graph in Proposition 4.10 is itself a path and has pathwidth one. Hence bounded pathwidth alone does not imply a  $2^{O(w)}$   $\text{poly}(N)$  bound on the number of exact continuation states. The proposition exhibits  $2^{N-1}$  distinct continuation classes, so a binary encoding of the current class needs at least  $N - 1$  bits in the worst case. This is a state-cardinality lower bound, not an exponential-space lower bound, and it does not assert that the scalar survival exponent is computationally hard.

By contrast,  $R_s$  is an  $(N - 1)$ -state weighted linear representation of the source-power survival series. Such a representation need not distinguish all deterministic continuation classes (Carlyle and Paz, 1971). A statistic sufficient for pressure can therefore be much smaller as a state set than the minimal machine needed to reproduce every repair decision.

**Relation to earlier processes.** The reduction of a two-colour disagreement process to the number of vertices of one colour is closely related to gambler’s-ruin reductions for discordant voting (Cooper et al., 2018). In that model a selected discordant edge is resolved by copying one endpoint, whereas here both endpoints are independently redrawn. The novelty claimed here is not the birth–death calculation by itself, but its simultaneous extension to every source-power operator, the all-power change-of-measure identity (102), the tape-dimension conclusions, and the separation between an exact pressure state and an exact continuation state. The instance is also nonextremal: adjacent bad events can occur together. Thus the extremal-instance analysis of partial rejection sampling does not yield this reduction (Guo et al., 2019; Jerrum, 2024).

**A second extremal benchmark.** Appendix D gives an adjacent-rise extremal family in which an  $(n + 1)$ -vertex tridiagonal pressure matrix replaces  $2^n$  exact continuation states. It is included as an application and benchmark for the source-power reduction, rather than as a separate principal theorem.

## 5 Graph summaries and computational limits

The exact regimes above use information not retained by graph and marginal summaries. The Lovász local lemma summarizes event interactions by a dependency graph, and many algorithmic proofs use that graph to organize repair histories. This section shows that the omitted information can change the exact tail and the geometry of the nonterminating tapes of a fixed computation.

Let  $\mathcal{A}$  be a finite family of bad events. In the variable model, the standard dependency graph  $D$  joins two events when their variable scopes overlap. Shearer’s region is the sharp worst-case criterion, based only on  $D$  and the marginal probabilities  $(p_A)$ , for guaranteeing that all bad events can be avoided (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005). Algorithmic local-lemma analyses use the same graph to organize and weigh repair histories (Moser and Tardos, 2010; Haeupler and Harris, 2017).

This compression of an event system is deliberately coarse. It records which repairs may interact, but not set inclusion, higher-order intersections, or the mass of  $\bigcup_A A$ . The following calculations compare the exact assignment-state description with graph-only bounds and then isolate the information lost by the graph summary.

### 5.1 Full-state formulas and graph bounds

Every finite variable-model resampling algorithm is a finite-state randomized computation on assignments. Set  $P = P_{\text{tape}}$ , the sequential source law of the initial assignment and subsequent resampling draws; finite initial factors do not affect the dimensions below. Delete zero-probability outcomes and assume that every surviving source outcome has probability strictly less than one. Fix a deterministic rule for selecting a currently true bad event. On reachable nonterminal assignments and for  $\theta \geq 0$ , define

$$(B_\theta)_{xy} = \sum_{z: F(x,z)=y} \pi_{v(A(x))}(z)^\theta, \quad (109)$$

where  $A(x)$  is the selected event and  $F(x, z)$  is the updated assignment. Decompose the nonterminal transition graph into reachable strongly connected components  $C$  containing a directed cycle, where a strongly connected component is a maximal set of states that are mutually reachable by directed paths. Let  $B_{\theta,C}$  be the restriction of  $B_\theta$  to  $C$ . If no such component exists, then  $\mathcal{N} = \emptyset$  and its dimension is zero. Otherwise, Proposition 4.1 gives

$$\dim_H^P \mathcal{N} = \max_C \delta_C, \quad \rho(B_{\delta_C, C}) = 1. \quad (110)$$

Here  $\delta_C \in [0, 1]$  is the unique solution of the displayed spectral-radius equation for component  $C$ . This formula is exact, but its matrix usually has exponentially many rows and columns.

**The graph bound.** Classical entropy-compression and Shearer arguments use the dependency graph and the probabilities  $p_A$  to bound the total weight of possible repair lists and traces (Moser and Tardos, 2010; Jerrum, 2024; Shearer, 1985; Scott and Sokal, 2005). Only the complete-graph

case is needed here. If  $D = K_m$  is complete on  $m = |\mathcal{A}|$  event vertices, the graph-only enumeration allows every word  $(A_1, \dots, A_t) \in \mathcal{A}^t$  and assigns it weight  $\prod_{i=1}^t p_{A_i}$ . Its total weight is

$$\left( \sum_{A \in \mathcal{A}} p_A \right)^t.$$

Put  $S = \sum_A p_A$  and choose the history law

$$Q_T(A_1, \dots, A_T) = \frac{\prod_{t=1}^T p_{A_t}}{S^T}$$

on all event words. Theorem 2.1 then gives  $\Delta_T = -T \log_2 S$ . If  $S < 1$ , it follows that

$$\mathbb{P}(E_T) \leq S^T. \quad (111)$$

The following dimension estimate requires one common fixed block. Let  $\Omega = [q]^b$  carry the uniform law, regard every bad event as a subset of this same block, and suppose that every repair redraws all  $b$  coordinates. Thus every repair consumes exactly  $b$   $q$ -ary symbols and  $|\mathcal{A}| = q^b p_A$ . Let  $P$  be the resulting product source on consumed blocks. Summing the compatible overwritten blocks over all event words gives at most

$$\sum_{(A_1, \dots, A_T)} \prod_{t=1}^T |A_t| = (q^b S)^T$$

length- $T$  overwritten-block strings. The final block contributes only the fixed factor  $q^b$ , which disappears after normalization. Coding the resulting prefixes gives

$$d_{\text{graph}} = \left\lceil 1 + \frac{\log_2 S}{b \log_2 q} \right\rceil_{[0,1]}, \quad \text{Dim}_{\text{eff}, P}(\omega) \leq d_{\text{graph}} \quad (\omega \in \mathcal{N}), \quad \dim_H^P \mathcal{N} \leq d_{\text{graph}}, \quad (112)$$

where  $[\cdot]_{[0,1]}$  denotes truncation to  $[0, 1]$ . These are graph-based upper bounds; they need not equal the survival probability or dimension for a fixed selection rule. Equation (112) is not asserted for unequal repair scopes. Padding shorter repairs would be legitimate only if the padding symbols were declared consumed tape coordinates and included in the metric; no such convention is used here. A short graph description also does not make exact computation easy: evaluating the relevant independence polynomial and testing membership in the general Shearer region are #P-hard in general (Harvey et al., 2018).

## 5.2 Whole-block separation

Let  $\Omega$  be a finite block space with source law  $\pi$  satisfying  $0 < \pi(x) < 1$  for every  $x$ , and let  $\mathcal{A}$  be any finite family of bad subsets. Every repair redraws the whole block from  $\pi$ . Fix a deterministic selector  $\phi$  that, for each  $x \in U = \bigcup_{A \in \mathcal{A}} A$ , chooses one event  $\phi(x)$  containing  $x$ . The selector partitions the bad union into the disjoint regions

$$C_A = \{x \in U : \phi(x) = A\}. \quad (113)$$

Let  $\mathcal{I} = \{A \in \mathcal{A} : C_A \neq \emptyset\}$  index the nonempty regions. For a laminar family with the inclusion-minimal selector,  $C_A$  is  $A$  minus its maximal proper subevents. Define

$$w_A(\theta) = \sum_{x \in C_A} \pi(x)^\theta, \quad W(\theta) = \sum_{A \in \mathcal{I}} w_A(\theta) = \sum_{x \in U} \pi(x)^\theta. \quad (114)$$

Write  $w(\theta) = (w_A(\theta))_{A \in \mathcal{I}}$  for the corresponding column vector.

**Theorem 5.1** (Exact whole-block calculation). *Assume  $U \neq \emptyset$ , and let  $P = \pi^{\mathbb{N}}$ . There is a unique  $\delta \in [0, 1]$  such that*

$$W(\delta) = \sum_{x \in U} \pi(x)^\delta = 1. \quad (115)$$

- (a) *If the finite source  $\pi$  and the set  $U$  are computable, then  $\delta$  and the law  $Q_\delta$  below are computable, and every nonterminating tape satisfies*

$$\text{Dim}_{\text{eff}, P}(\omega) \leq \delta \quad (\omega \in \mathcal{N} = U^{\mathbb{N}}). \quad (116)$$

*Every tape that is Martin–Löf random for  $Q_\delta^{\mathbb{N}}$  satisfies  $\dim_{\text{eff}, P}(\omega) = \text{Dim}_{\text{eff}, P}(\omega) = \delta$ . Moreover,*

$$\dim_H^P \mathcal{N} = \delta, \quad \mathbb{P}\{\tau \geq t\} = \pi(U)^t. \quad (117)$$

*Under the uniform law on  $\Omega = [q]^b$ , writing  $M = |U|$ ,*

$$\dim_H \mathcal{N} = \delta = \frac{\log_2 M}{b \log_2 q}, \quad \mathbb{P}\{\tau \geq t\} = \left(\frac{M}{q^b}\right)^t. \quad (118)$$

*In particular, under a uniform source, a tape of effective strong dimension greater than  $\delta$  must make the algorithm terminate.*

- (b) *Group rejected blocks according to the region  $C_A$  that contains them. For every  $\theta \geq 0$ , the resulting weighted transition matrix is*

$$\widehat{B}_\theta(A, B) = w_B(\theta) \quad (A, B \in \mathcal{I}). \quad (119)$$

*Its spectral radius is  $W(\theta)$ , the same as that of the full transition matrix, and the all-ones vector  $\mathbf{1} \in \mathbb{R}^{\mathcal{I}}$  is a right eigenvector.*

- (c) *The dimension and survival bounds are attained by the following reversible record laws. The dimension-attaining law is*

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}. \quad (120)$$

*For every  $T$ -repair execution, there is a probability record law  $G_{\delta, T}$  satisfying*

$$\log_2 \frac{G_{\delta, T}(S_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = (1 - \delta) \sum_{t=1}^T \log_2 \frac{1}{\pi(Y_t)}. \quad (121)$$

*The second is the source conditioned on another rejection,  $Q_{\text{surv}}(x) = \pi(x) \mathbf{1}\{x \in U\} / \pi(U)$ . The corresponding record law  $G_{\text{surv}, T}$  satisfies*

$$\log_2 \frac{G_{\text{surv}, T}(S_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = T \log_2 \frac{1}{\pi(U)}. \quad (122)$$

The proof is given in Appendix C.

**Remark 5.2** (The one-state reduction). Every repair redraws the whole block. Consequently, the next step is independent of the current rejected block, and the one-state matrix  $[W(\theta)]$  already gives both  $W(1) = \pi(U)$  and the dimension root. The disjoint regions retain the recorded reason for rejection, but do not change the stopping dynamics. Partially overlapping local repairs, which leave relevant state unchanged, are the harder case.

### 5.3 Succinct input

A one-state representation need not be easy to evaluate when its transition weight is given succinctly. Let  $C : \{0, 1\}^N \rightarrow \{0, 1\}$  be a Boolean circuit, put  $U_C = \{x : C(x) = 1\}$ , and assume  $1 \leq |U_C| < 2^N$ . Consider the sampler that draws independent uniform  $N$ -bit blocks and continues exactly on  $U_C$ . Let

$$\lambda(C) = \lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T}, \quad d(C) = \dim_H \mathcal{N}_C, \quad \mathcal{N}_C = U_C^{\mathbb{N}}.$$

Identify a block tape with its binary concatenation and use the standard binary metric on  $\{0, 1\}^{\mathbb{N}}$ .

**Proposition 5.3** (Hardness of succinct whole-block pressure). *Writing  $M_C = |U_C|$ , the one-state source-power operator and its exact quantities are*

$$R_s = \lfloor M_C 2^{-Ns} \rfloor, \quad \lambda(C) = \frac{M_C}{2^N}, \quad d(C) = \frac{\log_2 M_C}{N}. \quad (123)$$

Moreover,  $d(C)$  is the maximal effective strong dimension of a tape in  $\mathcal{N}_C$ . Even under the promise  $\lambda(C) < 1/3$ :

- (a) for every fixed  $0 < \varepsilon < 1/4$ , approximating  $d(C)$  to additive error at most  $\varepsilon$  is NP-hard;
- (b) computing the rational number  $\lambda(C)$  exactly is #P-hard under polynomial-time metric reductions.

The proof is in Appendix C. This baseline hardness lies in evaluating a single succinct transition weight; it is not a hardness theorem for bounded-scope overlapping repair.

The survival law is also a direct application of Theorem 2.1. Relabel each repair by the disjoint region  $C_A$  containing the overwritten block; this does not change the run or its resampling rule. Put  $r_A = \pi(C_A)$  and choose

$$Q_T(A_1, \dots, A_T) = \prod_{t=1}^T \frac{r_{A_t}}{\pi(U)}.$$

For the relabeled event  $C_{A_t}$ , the first term in (7) is  $\log_2(1/r_{A_t})$ , whereas encoding the region label under  $Q_T$  costs  $\log_2(\pi(U)/r_{A_t})$  bits. Their difference is  $\log_2(1/\pi(U))$  at every repair, giving (122). Conditioning only on the original event  $A_t$  would count overlapping mass more than once.

**Comparison with the dependency graph.** Under the standard variable-overlap definition, all common-block events are pairwise adjacent. The graph calculation therefore uses the one-step weight  $\sum_{A \in \mathcal{A}} \pi(A)$ , while the true rejection probability is  $\pi(U)$ . Under the uniform source on  $[q]^b$ , the resulting upper bound and the true dimension are

$$1 + \frac{\log_2 \sum_{A \in \mathcal{A}} \pi(A)}{b \log_2 q} \quad \text{and} \quad 1 + \frac{\log_2 \pi(U)}{b \log_2 q}, \quad (124)$$

again truncated to  $[0, 1]$ . Before truncation, the two expressions agree when the events are disjoint, whereas overlap produces overcounting. Truncation can conceal this difference when both values are at least one.

In the variable model, a nonedge means that the scopes are disjoint and the corresponding repairs do not interact; an edge marks only possible interaction. A complete graph therefore contains no information about how the corresponding bad events intersect.

**Example 5.4** (Pairwise data do not determine nontermination). Let the source be uniform on  $\Omega = \{0, 1\}^4$ . Begin with four subsets of  $\{0, 1\}^3$ ,

$$\begin{aligned} A_0 &= \{000, 001, 010\}, & B_0 &= \{000, 011, 100\}, \\ C_{\Delta,0} &= \{001, 011, 101\}, & C_{\star,0} &= \{000, 101, 110\}, \end{aligned}$$

and append a zero as the fourth coordinate:  $A = A_0 \times \{0\}$ ,  $B = B_0 \times \{0\}$ ,  $C_{\Delta} = C_{\Delta,0} \times \{0\}$ , and  $C_{\star} = C_{\star,0} \times \{0\}$ . Compare the whole-block systems

$$\mathcal{S}_{\Delta} = \{A, B, C_{\Delta}\}, \quad \mathcal{S}_{\star} = \{A, B, C_{\star}\}.$$

Each event depends essentially on all four coordinates: the displayed three-bit set depends on each of its coordinates, and flipping the appended zero to one changes membership. Thus both event–variable incidence graphs are  $K_{3,4}$  and both dependency graphs are  $K_3$ .

Every event has probability  $3/16$  in both systems. Moreover,

$$\begin{aligned} A \cap B &= \{0000\}, & A \cap C_{\Delta} &= \{0010\}, & B \cap C_{\Delta} &= \{0110\}, \\ A \cap B &= \{0000\}, & A \cap C_{\star} &= \{0000\}, & B \cap C_{\star} &= \{0000\}. \end{aligned}$$

Hence every pairwise intersection has probability  $1/16$  in both systems, whereas

$$A \cap B \cap C_{\Delta} = \emptyset, \quad A \cap B \cap C_{\star} = \{0000\}.$$

Inclusion–exclusion gives union sizes six and seven. Therefore, for every deterministic selector,

$$\mathbb{P}_{\Delta}\{\tau \geq t\} = \left(\frac{3}{8}\right)^t, \quad \mathbb{P}_{\star}\{\tau \geq t\} = \left(\frac{7}{16}\right)^t,$$

with survival exponents

$$\alpha_{\Delta} = \log_2 \frac{8}{3}, \quad \alpha_{\star} = \log_2 \frac{16}{7}.$$

Writing  $\mathcal{N}_{\Delta}$  and  $\mathcal{N}_{\star}$  for the two nonterminating sets, Theorem 5.1 gives

$$\max_{\omega \in \mathcal{N}_{\Delta}} \text{Dim}_{\text{eff}}(\omega) = \dim_H \mathcal{N}_{\Delta} = \frac{\log_2 6}{4}, \quad \max_{\omega \in \mathcal{N}_{\star}} \text{Dim}_{\text{eff}}(\omega) = \dim_H \mathcal{N}_{\star} = \frac{\log_2 7}{4}.$$

Thus the incidence graph, dependency graph, event marginals, and all pairwise intersection probabilities together determine neither the exact survival exponent nor either tape dimension. The two systems first differ at their triple intersection.

**Relation to the variable local lemma.** The separation does not contradict the tightness of the Shearer region, which is a worst-case statement over event systems with fixed graph and event probabilities. In fact, the common marginal vector in Example 5.4 satisfies

$$3 \cdot \frac{3}{16} = \frac{9}{16} < 1,$$

and therefore lies strictly inside the Shearer region for  $K_3$ . The variable local lemma gives avoidability guarantees from event probabilities and event–variable structure (He et al., 2017); algorithmic refinements give convergence bounds, including criteria that use intersection information (Moser and Tardos, 2010; He et al., 2023). Thus the loss of intersection information by dependency graphs is not itself new. The separation above concerns three different outputs of a fixed process: its survival

exponent, maximal effective strong dimension, and Hausdorff dimension. The backward identity supplies tail and individual-tape complexity bounds for every finite run in the model, whether or not a local-lemma criterion is satisfied. For the whole-block systems solved here, combining this identity with the product-set calculation determines the exact survival exponent and both dimensions. What differs in Example 5.4 is not local-lemma feasibility but the exact survival law and exceptional set of the specified whole-block process. This is a tape-level analysis of fixed computations, not a stronger universal local-lemma theorem.

## 6 Conclusion

Nontermination has two complementary descriptions. Reading one finite run backward gives an exact likelihood ratio: its tail bounds continued execution, and its coding form bounds the complexity of that tape. Summing the probabilities of all live cylinders to the power  $s$  gives a pressure: at  $s = 1$  it is survival, while its critical power measures the size of the whole exceptional set. The first description is pathwise; the second is setwise.

The central result makes the setwise description local. Whenever the powered kernels of nonadjacent repairs commute, the exact-prefix inequality bounds each repair history by a product of powered transition matrices. Summing over histories gives a parameter-wise trace/Shearer bound, uniformly over deterministic nonanticipating repair rules. At power one this bounds continued execution; below one it bounds weak-source survival and the two dimensions. For  $k$ -SAT the threshold is graph-specific: conditional min-entropy above  $\log_2 \kappa_D$  per redraw gives an exponential tail. Tree formulas attain the corresponding maximum-degree dimension bound asymptotically, while a clique gives the exact one-step threshold on its graph. The pointwise upper bound depends only on clauses repaired infinitely often.

The exact examples and hardness results mark the reach of this principle. Extremal partial rejection reaches a powered Shearer boundary; nonextremal disagreement repair has an exact linear-size reduction for every power; and whole-block systems show that pairwise event data can miss both survival and dimension. What remains open is equally precise. Tree sharpness concerns dimension and global tape domination, not the worst-case one-step weak source for nonextremal overlapping repairs. More broadly, the full assignment is still the only general exact state and is usually exponential. The main problem is to identify a natural causal or boundary width yielding a compact exact source-power representation, or to prove that no such representation can be recovered from a succinct process description.

## A Finite-state foundations and the one-state example

Let  $\mathcal{G}$  be a finite directed multigraph of nonterminal states with fixed initial state  $s_0$ . Each outgoing edge  $e : s \rightarrow t$  represents a distinct next source outcome and has probability  $p(e) > 0$ . The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path  $w = e_1 \cdots e_n$ , traversing states  $s_0, s_1, \dots, s_n$ , determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^n p(e_i).$$

Let  $\mathcal{W}_n$  be the set of surviving paths of length  $n$  from  $s_0$ . Assume that the reachable nonterminal graph is strongly connected, contains a directed cycle, and satisfies  $\max_e p(e) < 1$ . For  $\theta \geq 0$ , define

$$(B_\theta)_{st} = \sum_{e:s \rightarrow t} p(e)^\theta, \quad \Lambda(\theta) = \log_2 \rho(B_\theta). \quad (125)$$

For any square matrix  $C$ , its spectral radius is  $\rho(C) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } C\}$ . At  $\theta = 1$ ,  $B_1$  is the substochastic transition matrix among nonterminal states: each row sum is at most one, and the missing mass is the one-step termination probability. The quantity  $\Lambda(\theta)$  is the base-two pressure, the exponential growth rate of the total  $\theta$ -weight of surviving paths. Graphs that are not strongly connected are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension; if there is no such component, the nonterminating set is empty.

Use  $P[w]$  as the diameter of the cylinder  $[w]$ . The resulting Hausdorff dimension is the source-relative dimension  $\dim_H^P$  defined in the introduction; for a uniform  $q$ -ary source it is ordinary Hausdorff dimension in the  $q$ -adic metric. For an infinite path  $\omega = e_1 e_2 \cdots$ , use the corresponding pointwise definition

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(e_1 \cdots e_n)}{-\log_2 P[e_1 \cdots e_n]},$$

and define  $\dim_{\text{eff}, P}$  by replacing the limsup with a liminf.

**Proposition A.1** (Detailed finite-state formula). *There is a unique  $\delta \in [0, 1]$  satisfying  $\rho(B_\delta) = 1$ . Moreover,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \sum_{w \in \mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \quad (126)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \quad (127)$$

$$\dim_H^P \mathcal{N} = \delta. \quad (128)$$

If  $B_\delta h = h$  with  $h_s > 0$  for every state  $s$ , then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \quad (129)$$

is a probability law on surviving paths. For every  $w = e_1 \cdots e_n \in \mathcal{W}_n$ , define its probability under this law by

$$Q_\delta[w] = \prod_{i=1}^n Q_\delta(e_i \mid s_{i-1}).$$

Then it satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \quad (130)$$

For a computable system with computable  $Q_\delta$ , every nonterminating tape satisfies  $\text{Dim}_{\text{eff},P}(\omega) \leq \delta$ . A tape that is Martin–Löf random under  $Q_\delta$  satisfies  $\text{dim}_{\text{eff},P}(\omega) = \text{Dim}_{\text{eff},P}(\omega) = \delta$ .

The requirement that  $Q_\delta$  be computable is the effectivity condition needed for the pointwise conclusion. In the strict-interior finite cases below it follows automatically: if the edge probabilities are computable and

$$\rho(B_0) > 1 > \rho(B_1),$$

then  $\theta \mapsto \rho(B_\theta)$  is uniformly computable and strictly decreasing, so bisection computes  $\delta$ . Irreducibility makes the Perron eigenvalue simple; solving  $B_\delta h = h$  with  $\sum_s h_s = 1$  gives a computable positive eigenvector and hence a computable  $Q_\delta$ . Endpoint roots used below are explicit.

*Proof of Proposition A.1.* Since the graph contains a directed cycle,  $\rho(B_0) \geq 1$ . Since  $B_1$  is substochastic,  $\rho(B_1) \leq 1$ . If  $\theta' > \theta$ , then every nonzero entry contributed by an edge is smaller in  $B_{\theta'}$  than in  $B_\theta$ , because  $p(e) < 1$ . Strict Perron comparison on the common strongly connected support therefore gives  $\rho(B_{\theta'}) < \rho(B_\theta)$ . Together with continuity, this proves existence and uniqueness of  $\delta$ . The relevant matrix identity is

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1},$$

where  $e_{s_0}$  is the coordinate vector of the initial state and  $\mathbf{1}$  is the all-ones column vector. Perron–Frobenius theory proves (126); setting  $\theta = 1$  gives (127).

If  $\theta > \delta$ , then  $\rho(B_\theta) < 1$ , so the total  $\theta$ -weight of the length- $n$  surviving cylinders tends to zero. These cylinders cover  $\mathcal{N}$ , giving  $\text{dim}_H^P \mathcal{N} \leq \delta$ . At  $\theta = \delta$ , (129) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because  $h$  is positive on a finite state space, there is a constant  $C < \infty$ , independent of  $w$ , such that  $Q_\delta[w] \leq CP[w]^\delta$ . The mass-distribution principle states that a probability measure satisfying this cylinder bound forces  $\text{dim}_H^P \mathcal{N} \geq \delta$ . It therefore gives the required lower bound, proving (128) and (130) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system with computable  $Q_\delta$ , coding under  $Q_\delta$  gives, for every surviving prefix  $w$  of length  $n$ ,

$$K(w) \leq -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a  $Q_\delta$ -Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015).  $\square$

Under a uniform  $q$ -ary source, let  $M = (M_{st})$ , where  $M_{st}$  counts the source symbols that move  $s$  to  $t$ . Then  $B_\theta = q^{-\theta} M$ . If

$$\alpha = -\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}$$

is the survival exponent, then

$$\text{dim}_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Writing  $h_{\text{surv}} = \log_2 \rho(M)$  for the topological entropy, in bits per step, of the surviving path space gives

$$\alpha = \log_2 q - h_{\text{surv}}, \quad \dim_H \mathcal{N} = \frac{h_{\text{surv}}}{\log_2 q}.$$

Thus  $\alpha$  is the entropy deficit imposed by survival, not an entropy function. Equation (130) is the finite-state form of the comparison used in Theorem 2.1.

**Rejection sampling.** Let  $p = \pi(U)$ , let  $\nu = \pi(\cdot | U)$  be the law of a rejected symbol, and put  $\alpha = \log_2(1/p)$ .

**Proposition A.2** (Entropy bounds for the rejection dimension). *Assume  $0 < p < 1$ . If  $|U| = 1$ , then  $\delta = 0$ . If  $|U| \geq 2$ , define the Rényi entropy of order  $s \in (0, 1)$  by*

$$H_s(\nu) = \frac{1}{1-s} \log_2 \sum_{x \in U} \nu(x)^s,$$

and let  $H(\nu)$  be the Shannon entropy (Rényi, 1961; Shannon, 1948). Then

$$\delta = \frac{H_\delta(\nu)}{\alpha + H_\delta(\nu)} \tag{131}$$

and hence

$$\frac{H(\nu)}{\alpha + H(\nu)} \leq \delta \leq \frac{\log_2 |U|}{\alpha + \log_2 |U|}. \tag{132}$$

The two bounds coincide when  $\nu$  is uniform on  $U$ .

*Proof of Proposition A.2.* For  $|U| \geq 2$ , the root lies in  $(0, 1)$ . Since  $\pi(x) = p\nu(x)$  on  $U$ , the defining equation for  $\delta$  becomes

$$1 = p^\delta \sum_{x \in U} \nu(x)^\delta.$$

Taking logarithms gives  $\delta\alpha = (1 - \delta)H_\delta(\nu)$ , which proves (131). The bounds follow from  $H(\nu) \leq H_\delta(\nu) \leq \log_2 |U|$ .  $\square$

## B Likelihood and powered local repair

### Backward likelihood.

*Proof of Theorem 2.1.* Take the ratio of (6) and (5). The final-state factor cancels, as do all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(\Xi_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (7).

It remains to use this likelihood ratio. Let  $\mathcal{W}_T$  be the prefix-free set of minimal tape prefixes that produce at least  $T$  repairs, and let  $\mathcal{R}_T$  be the set of formal records on which  $G_T$  is defined. For  $w \in \mathcal{W}_T$ , write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)).$$

The reconstruction argument shows that  $R_T$  is injective.

First observe that  $G_T$  has total mass at most one. Indeed, for a fixed list  $\ell = (A_1, \dots, A_T)$ , the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \leq 1.$$

Define the corresponding mass on live tape prefixes by

$$\mu_T(w) = G_T(R_T(w)), \quad w \in \mathcal{W}_T.$$

Injectivity of  $R_T$  gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \leq \sum_{r \in \mathcal{R}_T} G_T(r) \leq 1.$$

Thus  $\mu_T$  is a subprobability mass function. Its total mass can be strictly less than one because  $Q_T$  may itself have mass less than one, or because  $G_T$  may assign mass to formal records that are not produced by the algorithm.

For every  $w \in \mathcal{W}_T$ , the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because  $\Delta_T = -\infty$  outside  $E_T$  and the cylinders determined by  $\mathcal{W}_T$  are disjoint, for every  $u \geq 0$ ,

$$\begin{aligned} \mathbb{P}\{\Delta_T \geq u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} P_{\text{tape}}(w) \\ &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} 2^{-\Delta_T(w)} \mu_T(w) \\ &\leq 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} \mu_T(w) \\ &\leq 2^{-u}. \end{aligned}$$

This proves (8); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Use the canonical effective encoding of the consumed-choice tree from Section 2.1, and extend  $\mu_T$  by zero off  $\mathcal{W}_T$ . Membership in  $\mathcal{W}_T$  is decidable uniformly from  $(T, w)$  by simulating the deterministic computation, and  $w \mapsto R_T(w)$  is uniformly computable. The finite source factors and  $Q_T$  are uniformly computable by hypothesis. Hence  $(\mu_T)_{T \geq 1}$  is a uniformly lower semicomputable family of subprobability masses (indeed, uniformly computable). The Kraft–Chaitin coding theorem for lower semicomputable semimeasures, equivalently Shannon–Fano coding in this discrete setting (Shannon, 1948; Fano, 1949; Li and Vitányi, 2019), gives

$$K(\Xi_T \mid T, \text{instance}) \leq -\log_2 \mu_T(\Xi_T) + O(1).$$

The constant is uniform in  $T$  and in the realized run. Since

$$-\log_2 \mu_T(\Xi_T) = -\log_2 P_{\text{tape}}(\Xi_T) - \Delta_T,$$

rearranging proves (9). □

*Proof of Corollary 2.2.* On  $\mathcal{W}_T$ ,

$$\Delta_T(w) = \log_2 \frac{m_T \nu_T(w)}{p_T P_T(w)}.$$

Taking  $P_T$ -expectations gives (10). For the second claim,  $\Delta_T = J_T + \log_2 Q_T(L_T)$ . If  $r_T$  is the conditional law of  $L_T$  given  $E_T$ , then

$$\sum_{\ell} r_T(\ell) \log_2 Q_T(\ell) = -H_2(r_T) - D_2(r_T \| Q_T)$$

when  $Q_T$  has total mass one; allocating less mass can only decrease the left side. The maximum is attained at  $Q_T = r_T$ .  $\square$

*Proof of Corollary 2.4.* The assumptions give  $\Delta_T \geq (\beta - \gamma)T - (C_0 + C_1)$ . Equation (15) follows from (8). Equation (9), together with the  $O(\log_2 T)$  bits needed to specify  $T$ , gives  $K(\Xi_T) \leq c_0 + cT - (\beta - \gamma)T + o(T)$  along an infinite execution. If  $\beta - \gamma > c$ , this is impossible for all large  $T$ , so no infinite run exists. Otherwise divide by  $c_0 + cT$ ; the gap between consecutive repair prefixes is the fixed number  $c$  of bits.  $\square$

### Powered matrix theorem.

*Proof of Lemma 3.1.* The  $(\sigma, \tau)$  entry of  $K_{f,s}K_{g,s}$  is the sum over all action-labelled two-step paths

$$\sigma \xrightarrow{f,a} \sigma' \xrightarrow{g,b} \tau$$

of

$$(\rho_f(a \mid \sigma) \rho_g(b \mid \sigma'))^s.$$

The diamond bijection maps these paths onto the corresponding  $g, f$  paths with the same endpoints and preserves each product before it is powered. The two matrix entries are therefore equal.

For the fixed-arity statement, let  $m_f(\sigma, \tau)$  be the number of action labels that move  $\sigma$  to  $\tau$  when  $f$  is addressed. If  $f$  is present at  $\sigma$ , then

$$K_{f,s}(\sigma, \tau) = m_f(\sigma, \tau) M_f^{-s} = M_f^{1-s} K_{f,1}(\sigma, \tau).$$

Both sides vanish if  $f$  is absent. Hence, for nonadjacent  $f, g$  whose ordinary matrices commute,

$$K_{f,s}K_{g,s} = M_f^{1-s} M_g^{1-s} K_{f,1}K_{g,1} = K_{g,s}K_{f,s}.$$

$\square$

*Proof of Lemma 3.2.* First observe that  $K_{H,s}$  is well defined. If  $x$  and  $y$  are two sources of  $H$ , their labels are nonadjacent; otherwise  $H$  would contain an edge between them. Their powered matrices commute, and induction on  $|H|$  shows that deleting the sources in either order gives the same product.

A stronger conditional statement is proved by induction. Fix an arbitrary past history  $\mathfrak{h}$  ending at state  $\sigma$ , and let  $M_{\mathfrak{h}}(H)$  be the powered mass of all continuations whose next  $|H|$  repairs induce  $H$ . The claim is

$$M_{\mathfrak{h}}(H) \leq e_{\sigma}^{\top} K_{H,s} \mathbf{1}. \quad (133)$$

It is immediate for  $H = \emptyset$ .

Suppose  $H \neq \emptyset$ , and let  $f$  be the flaw selected after  $\mathfrak{h}$ . If no source of  $H$  is labelled  $f$ , then  $M_{\mathfrak{h}}(H) = 0$ . Otherwise there is a unique such source  $x$ , since two vertices with the same label are adjacent and cannot both be sources. After an action  $a \in \mathcal{A}_f(\sigma)$  leading to  $\tau = \Phi_f(\sigma, a)$ , the remaining repairs must induce  $H - x$ . Applying the induction hypothesis separately after every possible action gives

$$\begin{aligned} M_{\mathfrak{h}}(H) &\leq \sum_{a \in \mathcal{A}_f(\sigma)} \rho_f(a \mid \sigma)^s e_{\Phi_f(\sigma, a)}^{\top} K_{H-x, s} \mathbf{1} \\ &= e_{\sigma}^{\top} K_{f, s} K_{H-x, s} \mathbf{1} = e_{\sigma}^{\top} K_{H, s} \mathbf{1}. \end{aligned}$$

Actions leading to the same state are added in  $K_{f, s}$ ; no recovery of an action from its endpoints is used. The induction is uniform over  $\mathfrak{h}$ , so the selector may use the complete past action history.

Finally, sum (133) over the initial source symbols. Their powered masses at state  $\sigma$  sum to  $b_s(\sigma)$ , which gives (28).  $\square$

*Proof of Theorem 3.3.* For every full history DAG  $H$ , the definition of  $\lambda_f(s)$  gives

$$v_s^{\top} K_{H, s} \mathbf{1} \leq \left( \prod_{x \in H} \lambda_{L(x)}(s) \right) v_s^{\top} \mathbf{1} = \prod_{x \in H} \lambda_{L(x)}(s).$$

Since  $b_s \leq \gamma_s v_s$  coordinatewise, Lemma 3.2 yields

$$\sum_{w \in \mathcal{L}_T^{\Lambda}(H)} P_{\text{tape}}[w]^s \leq \gamma_s \prod_{x \in H} \lambda_{L(x)}(s). \quad (134)$$

The full history DAG of a flaw word is its labelled dependence graph. Full history DAGs, understood up to the label-preserving isomorphism fixed above, are therefore in one-to-one correspondence with traces; only a subset need be realizable by the algorithm. Summing (134) over realized DAGs and then enlarging to all traces of length  $T$  proves

$$Z_T^{\Lambda}(s) \leq \gamma_s \sum_{\substack{h \in \mathcal{M}(D) \\ |h|=T}} \prod_f \lambda_f(s)^{N_f(h)} = \gamma_s a_T(s).$$

The Cartier–Foata identity gives the parameter-wise formal-power-series bound.

If  $q\lambda(s)$  lies in the open Shearer region, the series in (27) converges at  $z = q$  and has nonnegative parameters. Hence

$$q^T a_T(s) \leq \sum_{t \geq 0} q^t a_t(s) = \frac{1}{P_D(q\lambda(s))},$$

which proves (32).

It remains to derive the local cluster estimate. Assume first that  $0 < \theta_s < 1$  and set

$$\bar{\lambda}_f = \frac{\lambda_f(s)}{\theta_s}.$$

By definition of  $\theta_s$ ,

$$\bar{\lambda}_f \sum_{I \in \text{Ind}(\Gamma(f))} \eta(I) \leq \eta_f.$$

The restricted cluster-expansion enumeration therefore gives, for every independent sink set  $R$ ,

$$\sum_{\substack{h \in \mathcal{M}(D) \\ \text{sink}(h)=R}} \prod_f \bar{\lambda}_f^{N_f(h)} \leq \eta(R). \quad (135)$$

This is the standard trace-DAG form of the cluster bound; see [Harris et al. \(2025, Proposition 3.9\)](#). Restricting the sum to traces of length  $T$  and restoring the factor  $\theta_s^T$  gives

$$a_T(s) \leq \theta_s^T \sum_{R \in \text{Ind}(F)} \eta(R).$$

Together with (30), this proves (33). If  $\theta_s = 0$ , then every  $\lambda_f(s) = 0$ , so the assertion is immediate for  $T \geq 1$ ; at  $T = 0$  it follows from  $\sum_{\sigma} b_s(\sigma) \leq \gamma_s$ .  $\square$

*Proof of Corollary 3.4.* The depth- $T$  cylinders indexed by  $\mathcal{L}_T^\Lambda$  cover  $\mathcal{N}$  and have source diameters at most a fixed initial factor times  $\rho_{\max}^T$ . By (34),

$$\sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s \leq C_s \vartheta_s^T \longrightarrow 0.$$

This proves that the source-relative  $s$ -dimensional Hausdorff measure of  $\mathcal{N}$  is zero.

For the effective statement, define on  $\mathcal{L}_T^\Lambda$

$$\kappa_{T,s}(w) = \frac{P_{\text{tape}}[w]^s}{\widehat{C}_s \widehat{\vartheta}_s^T}.$$

These are subprobability masses because

$$\sum_{w \in \mathcal{L}_T^\Lambda} \kappa_{T,s}(w) \leq \frac{C_s \vartheta_s^T}{\widehat{C}_s \widehat{\vartheta}_s^T} \leq 1.$$

They are uniformly computable under the stated assumptions. Prefix-free coding gives

$$\begin{aligned} K(w \mid T, \text{instance}) &\leq -\log_2 \kappa_{T,s}(w) + O(1) \\ &= s \log_2 \frac{1}{P_{\text{tape}}[w]} + \log_2 \widehat{C}_s + T \log_2 \widehat{\vartheta}_s + O(1). \end{aligned}$$

Removing the condition  $T$  costs  $O(\log T)$ . Since  $\rho_{\max} < 1$ , the source self-information of a depth- $T$  prefix grows at least linearly in  $T$ . Divide by this self-information and take a limsup; the term  $T \log_2 \widehat{\vartheta}_s$  is nonpositive, and the result follows.  $\square$

*Proof of Corollary 3.5.* The cylinders indexed by  $\mathcal{L}_T^\Lambda$  are disjoint. Hence

$$\mathbb{P}_\nu\{\tau \geq T\} = \sum_{w \in \mathcal{L}_T^\Lambda} \nu[w] \leq D \sum_{w \in \mathcal{L}_T^\Lambda} P_{\text{tape}}[w]^s.$$

Apply (34).  $\square$

**Weak-randomness applications.** For the variable-model calculation, take

$$v_s(\sigma) = \frac{\pi(\sigma)^s}{\sum_{\xi \in \Omega} \pi(\xi)^s}.$$

For  $S = v(A)$  and a fixed terminal assignment  $\tau$ ,

$$\begin{aligned} \sum_{\sigma} v_s(\sigma) K_{A,s}(\sigma, \tau) &= v_s(\tau) \sum_{b \in A} \pi_S(b)^s \\ &= v_s(\tau) r_A(s). \end{aligned}$$

This proves (40).

*Proof of Theorem 3.6.* For uniform  $k$ -clause resampling,  $\lambda_A(s) = 2^{-ks}$  for every clause  $A$ . A fixed initial assignment and the uniform choice of  $v_s$  give  $\gamma_s = 2^n$ . Therefore Theorem 3.3 gives

$$Z_T^\Lambda(s) \leq 2^n 2^{-ksT} |\{h \in \mathcal{M}(D) : |h| = T\}|.$$

For every  $\xi > \kappa_D$ , the Cartier–Foata series at  $z = \xi^{-1}$  gives

$$|\{h \in \mathcal{M}(D) : |h| = T\}| \leq \frac{\xi^T}{P_D(\xi^{-1} \mathbf{1})}.$$

Multiplying by  $D_0$  proves (43). A conditional atom bound implies  $\nu[w] \leq 2^{-hT} = P[w]^{h/k}$  by the chain rule, so this is the special case  $s = h/k$  and  $D_0 = 1$ .

For every  $s > \log_2(\kappa_D)/k$ , choose  $\xi \in (\kappa_D, 2^{ks})$ . The resulting source-power sum decays exponentially, which proves the Hausdorff and, under computability, effective dimension bounds by Corollary 3.4. Finally, the symmetric form of Shearer’s theorem (Shearer, 1985; Scott and Sokal, 2005) states that a graph of maximum degree  $d \geq 2$  has  $z\mathbf{1}$  in its open Shearer region whenever

$$0 \leq z < \frac{(d-1)^{d-1}}{d^d}.$$

Hence  $R_D \geq (d-1)^{d-1}/d^d$  and  $\kappa_D \leq c_d$ . □

*Proof of Corollary 3.7.* Fix a nonempty  $R \subseteq F$ . If  $\text{Rec}(\omega) = R$ , every clause outside  $R$  is repaired only finitely often. After a finite tape prefix, the run therefore repairs only clauses in  $R$ . Fix such a prefix  $u$ . On a history of the instance that retains only the clauses in  $R$ , prepend  $u$  and apply the original selector whenever it chooses a currently violated clause in  $R$ ; otherwise choose the first currently violated clause of  $R$  in a fixed order. This defines a deterministic nonanticipating selector for the reduced instance and agrees with the original run on every continuation after  $u$  whose later repair labels all lie in  $R$ . It is computable whenever the original selector is. The proof of Theorem 3.6 now applies to the induced graph  $D[R]$ . For every  $s > \log_2(\kappa_{D[R]})/k$ , the source-power sum of the suffix decays exponentially.

There are countably many finite prefixes after which the run remains in  $R$ . Prefixing rescales the source metric by a constant, and Hausdorff dimension is stable under countable unions. This proves the setwise bound. In the computable case, a fixed prefix and the finite set  $R$  cost only an additive constant in a program description. Prefix-free coding of the same suffix bound and passage to the infimum over  $s$  prove (48). □

*Proof of Theorem 3.8.* For a violated  $k$ -clause under the uniform reference source, precisely one local assignment is bad, so its powered charge is  $2^{-ks}$ . Take  $v_s$  uniform on the  $2^n$  assignments. A fixed initial assignment gives  $\gamma_s = 2^n$ . Set  $\eta_A = \eta = 1/(d-1)$ . A clause is adjacent to every other member of its inclusive neighborhood. An independent subset of that neighborhood therefore either consists of the clause itself or is contained in its at most  $d$  neighbors. Hence

$$\sum_{I \in \text{Ind}(\Gamma(A))} \eta(I) \leq \eta + (1 + \eta)^d.$$

Consequently

$$\theta_s \leq 2^{-ks} \frac{\eta + (1 + \eta)^d}{\eta} = 2^{-ks} \left( 1 + \frac{d^d}{(d-1)^{d-1}} \right) = 2^{-ks} A_d,$$

whereas

$$\sum_{R \in \text{Ind}(F)} \eta(R) \leq (1 + \eta)^m = \left( \frac{d}{d-1} \right)^m.$$

The atom condition implies that every length- $T$  redraw word has probability at most  $2^{-hT}$ . With  $s = h/k$ , this equals its uniform reference probability raised to  $s$ . If  $h = \log_2 A_d + \varepsilon$ , apply Corollary 3.5 to obtain (51).

Taking  $h = k$  gives (52). For the dimension statement, every  $s > \log_2 A_d/k$  makes  $\theta_s < 1$ . Corollary 3.4 applies for every such  $s$ ; let  $s$  decrease to the displayed value. The bound by one is automatic.  $\square$

*Proof of Corollary 3.9.* Conditioned on the past, injectivity of  $E_t$  makes every supported redraw have probability exactly  $2^{-r}$ . The encoder is chosen before  $U_t$ , so its dependence on the past does not change this fact. Apply Theorem 3.8 with  $h = r$ . Summing the tail by (64) gives (54), and every repair consumes exactly  $r$  bits.

If  $A_d \leq 2^{k-1}$  and  $r = \lceil \log_2 A_d + 1 \rceil$ , then  $r \leq k$ ,  $\varepsilon_r \geq 1$ , and  $r = O(\log(d+1))$ . Moreover,  $\log_2(d/(d-1)) = O(1/d)$  for  $d \geq 2$ . Substitution in (54) proves (55).  $\square$

*Proof of Corollary 3.10.* Condition on the entire past before a block is read. The chain rule and the Santha–Vazirani inequalities give, for every  $z \in \{0, 1\}^r$ ,

$$\mathbb{P}\{(U_{j+1}, \dots, U_{j+r}) = z \mid U_1, \dots, U_j\} \leq (1 - \alpha)^r.$$

After an injective map, the same upper bound holds for every redraw atom. Thus Theorem 3.8 applies with  $h = r \log_2(1/(1 - \alpha))$ .  $\square$

*Proof of Proposition 3.11.* Put  $\ell = \lceil \log_2(d+1) \rceil$  and choose distinct words  $u_1, \dots, u_{d+1} \in \{0, 1\}^\ell$ . Use  $\ell$  common variables and, for each  $i$ , a disjoint private block  $P_i$  of  $k - \ell$  variables. Let  $C_i$  be the unique clause on the common variables and  $P_i$  that is false when the common word is  $u_i$  and  $P_i$  is all zero. The clauses share the common variables, so their dependency graph is  $K_{d+1}$ , but their scopes are not equal. Distinct clauses cannot be false simultaneously. The formula is satisfiable, for example by setting one private variable in each  $P_i$  to one.

Start with every private block zero and the common word equal to  $u_1$ . Call  $i$  active when  $P_i$  is still zero and let  $q$  be the number of active indices. In every live state the common word is  $u_i$  for a unique active  $i$ , so  $C_i$  is the only violated clause. Write  $M = 2^{k-\ell}$ . When  $C_i$  is repaired, exactly  $q$  of its  $2^k$  redraws leave  $q$  unchanged: the new private block is zero and the new common word is  $u_j$  for an active  $j$ . Exactly  $(q-1)(M-1)$  redraws move to  $q-1$ : the new private block is nonzero

and the new common word is  $u_j$  for one of the other active indices. Every other redraw terminates. Thus the all-power live matrix on  $q = 1, \dots, d+1$  is bidiagonal, with

$$R_s(q, q) = q 2^{-ks}, \quad R_s(q, q-1) = (q-1)(M-1)2^{-ks}. \quad (136)$$

Its spectral radius is  $(d+1)2^{-ks}$ . Proposition 4.1 therefore gives (56) and (57), including pointwise attainment.

At  $q = d+1$ , instead redraw the selected scope uniformly from the  $d+1$  assignments  $(u_j, 0)$ ,  $1 \leq j \leq d+1$ . This source has conditional block min-entropy  $\log_2(d+1)$  and remains at  $q = d+1$  forever. If  $d+1 = 2^r$ , the redraw is the image of an injective map applied to  $r$  fresh fair bits. Finally,

$$\frac{c_d}{d+1} = \frac{d}{d+1} \left(1 + \frac{1}{d-1}\right)^{d-1} < e,$$

which proves the stated gap.  $\square$

*Proof of Theorem 3.12.* Orient every edge of  $T_{d,\ell}$  away from the root and introduce one Boolean variable  $X_e$  for each edge. For a vertex  $v$ , form a clause  $C_v$  containing one literal for each incident edge, using opposite signs at the two endpoints. Add  $k - \deg(v)$  private variables to  $C_v$ , each used in no other clause. This is possible because  $\deg(v) \leq d \leq k$ . Every clause has exactly  $k$  distinct variables, and every variable occurs once or twice.

Two clauses share a variable exactly when their vertices are adjacent. Their bad events prescribe opposite values to the shared edge variable and are therefore disjoint; nonadjacent clauses have disjoint scopes. The formula is satisfiable. For example, use the sign convention in which the parent literal on an edge is  $X_e$  and the child literal is  $\neg X_e$ , set every  $X_e = 0$ , and set one private literal of the root to true. Every nonroot clause is satisfied by its parent-edge literal, while the root has a private literal because its degree is  $d-1 < k$ .

Every bad event is atomic and has probability  $2^{-k}$ . The extremal source-power formula therefore gives (58) and (59), including effective attainment under computability.

It remains to calculate the trace growth. Put  $b = d-1$  and define

$$r_0(z) = z, \quad r_{j+1}(z) = \frac{z}{(1 - r_j(z))^b}.$$

As long as the denominators are positive,  $r_j(z)$  is the signed root-occupation ratio for a full  $b$ -ary subtree of height  $j$ . The first positive zero  $z_{d,\ell}$  of  $P_{T_{d,\ell}}(z\mathbf{1})$  is characterized by  $r_\ell(z_{d,\ell}) = 1$ . The critical value for the iteration is

$$z_* = \max_{0 \leq r < 1} r(1-r)^b = \frac{b^b}{(b+1)^{b+1}} = \frac{(d-1)^{d-1}}{d^d}.$$

If  $z < z_*$ , the iteration increases to the smaller fixed point of  $r = z/(1-r)^b$  and remains below one. If  $z > z_*$ , there is no fixed point in  $[0, 1)$ , so the iteration reaches one after finitely many steps. Hence  $z_{d,\ell} \downarrow z_*$  and

$$\kappa_{T_{d,\ell}} = z_{d,\ell}^{-1} \uparrow z_*^{-1} = c_d.$$

Finally put  $\delta_\ell = \log_2(\kappa_{T_{d,\ell}})/k$ . For the fixed selector, consider the finite live chain on full assignments. The exact growth calculation above and the finite-state decomposition give a reachable irreducible component  $C$  whose powered transition matrix satisfies  $\rho(B_{\delta_\ell, C}) = 1$ . Choose a positive right Perron vector  $h$  and a live prefix  $u$  entering  $C$ . On an internal edge  $e : x \rightarrow y$  of  $C$ , set

$$Q(e \mid x) = P(e)^{\delta_\ell} \frac{h(y)}{h(x)}.$$

Its row sums are one, so prefixing  $Q$  by  $u$  defines a probability law  $\nu_{\delta_\ell}$  supported on  $\mathcal{N}$ . Telescoping along extensions of  $u$ , and absorbing the source probability of  $u$  and the finitely many ratios of entries of  $h$  into  $D_\ell$ , gives (61) for every finite prefix  $w$ . Since  $P[w] \leq 1$ , the same inequality holds with  $P[w]^s$  for every  $0 \leq s \leq \delta_\ell$ .  $\square$

*Proof of Corollary 3.13.* For the matching oracle of Kolmogorov (2018), every flaw has

$$A = (2n - 1)(2n - 3)$$

equiprobable actions, distinct oracle executions have distinct outputs in the action-expanded graph, and the oracle is strongly commutative. Thus its powered charge is  $A^{-s}$ . Put  $C = (2n - 1)(q - 1)$ . The neighborhood of a flaw is covered by four families, one for each endpoint, each containing at most  $C$  flaws. Therefore, for constant  $\eta$ ,

$$\sum_{I \in \text{Ind}(\Gamma(f))} \eta^{|I|} \leq (1 + C\eta)^4.$$

Taking  $\eta = 1/(3C)$  gives

$$\theta_s \leq \frac{A^{-s}}{\eta} (1 + C\eta)^4 = \frac{256}{27} C A^{-s}.$$

There are at most

$$|F| \leq \frac{n(2n - 1)(q - 1)}{2}$$

flaws, and hence

$$\sum_{R \in \text{Ind}(F)} \eta^{|R|} \leq (1 + \eta)^{|F|} \leq e^{|F|\eta} \leq e^{n/6}.$$

With  $v_s$  uniform on the  $(2n - 1)!!$  perfect matchings, a fixed initial matching has  $\gamma_s = (2n - 1)!!$ . The conditional atom bound and the uniform reference oracle give power domination with  $s = h / \log_2 A$ . Corollary 3.5 now proves (62). Letting  $s$  decrease to  $\log_2((256/27)C) / \log_2 A$  proves the dimension bound. Under uniform actions, the strict condition  $\theta_1 < 1$  is equivalent to  $q - 1 < 27(2n - 3)/256$ .  $\square$

## C Whole-block systems and succinct input

### Whole-block resampling.

*Proof of Theorem 5.1.* The function  $W$  is continuous and strictly decreasing because  $0 < \pi(x) < 1$ . Moreover,  $W(0) = |U| \geq 1$  and  $W(1) = \pi(U) \leq 1$ , so (115) has a unique solution in  $[0, 1]$ .

The effectivity statement is direct. If  $|U| = 1$ , then  $\delta = 0$ ; if  $U = \Omega$ , then  $\delta = 1$ . Otherwise  $W(0) > 1 > W(1)$  and  $W$  is computable and strictly decreasing, so effective bisection computes  $\delta$ . The finite law  $Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}$  is therefore computable.

For full nonterminal states  $x, y \in U$ , resampling the common scope gives  $B_\theta(x, y) = \pi(y)^\theta$ . Hence, for every  $x \in C_A$ ,

$$\sum_{y \in C_B} B_\theta(x, y) = \sum_{y \in C_B} \pi(y)^\theta = w_B(\theta),$$

independently of  $x$ . Thus grouping states by the sets  $C_A$  leaves the weighted transition calculation unchanged and proves (119). The reduced matrix is  $\mathbf{1} w(\theta)^\top$ , so its only nonzero eigenvalue is  $W(\theta)$  and  $\mathbf{1}$  is a right eigenvector. Proposition 4.1 now gives  $\dim_H^P \mathcal{N} = \delta$ . Each repair draws a fresh block from  $\pi$  and continues precisely on  $U$ , proving the survival formula and its uniform specialization.

For the dimension bound, observe that  $Q_\delta$  is computable and supported on  $U$ . Every  $\omega \in U^\mathbb{N}$  therefore satisfies

$$K(\omega_{1:n}) \leq -\log_2 Q_\delta^\mathbb{N}[\omega_{1:n}] + O(\log_2 n) = \delta(-\log_2 P[\omega_{1:n}]) + O(\log_2 n).$$

Division by the source information proves (116). If  $\omega$  is Martin–Löf random for  $Q_\delta^\mathbb{N}$ , Levin–Schnorr gives the reverse inequality up to  $O(1)$ , and hence equality. This also proves (67).

It remains to construct the record laws. The repair list is determined by the overwritten states because  $A_t = \phi(Y_t)$ . Reversal recovers the used tape from  $(S_T, Y_{1:T})$ . The source factorization is

$$P_{\text{tape}}(\Xi_T) = \pi(S_T) \prod_{t=1}^T \pi(Y_t).$$

Since (115) holds, the record law

$$G_{\delta,T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^T \left[ \pi(y_t)^\delta \mathbf{1}\{y_t \in U\} \right]$$

has total mass one. Dividing by the source factorization proves (121). Similarly, define

$$G_{\text{surv},T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^T \frac{\pi(y_t) \mathbf{1}\{y_t \in U\}}{\pi(U)}.$$

This is a probability law, and division by the source factorization proves (122). The region containing  $y_t$  determines the recorded rejection label  $A_t = \phi(y_t)$ .  $\square$

### Succinct one-state hardness.

*Proof of Proposition 5.3.* Each surviving block has probability  $2^{-N}$  and there are  $M_C$  such blocks. Summing their  $s$ th powers gives  $R_s = [M_C 2^{-Ns}]$ . Therefore

$$\mathbb{P}\{\tau \geq T\} = \left( \frac{M_C}{2^N} \right)^T, \quad \dim_H \mathcal{N}_C = \frac{\log_2 M_C}{N}.$$

Theorem 5.1 bounds the effective strong dimension of every tape in  $\mathcal{N}_C$  by this value, while the computable uniform law on  $U_C$  has Martin–Löf-random tapes attaining it.

For the reduction, let  $\varphi$  be a Boolean formula on  $n \geq 1$  variables. Put  $m = n + 3$ ,  $N = 1 + n + m = 2n + 4$ , and write a block as  $(b, x, y) \in \{0, 1\} \times \{0, 1\}^n \times \{0, 1\}^m$ . Construct a circuit  $C_\varphi$  that accepts exactly when either

$$(b, x, y) = 0^N,$$

or

$$b = 1, \quad \varphi(x) = 1, \quad y_1 \oplus \cdots \oplus y_m = 0.$$

The cases are disjoint. If  $S = \#\{x : \varphi(x) = 1\}$ , then

$$|U_{C_\varphi}| = 1 + 2^{m-1}S. \tag{137}$$

Moreover,

$$\lambda(C_\varphi) \leq \frac{1 + 2^{m-1+n}}{2^N} = \frac{1}{4} + 2^{-N} \leq \frac{17}{64} < \frac{1}{3}.$$

If  $\varphi$  is unsatisfiable, then  $d(C_\varphi) = 0$ . If it is satisfiable, then

$$|U_{C_\varphi}| > 2^{m-1} = 2^{n+2} = 2^{N/2},$$

and hence  $d(C_\varphi) > 1/2$ . An additive- $\varepsilon$  approximation with  $\varepsilon < 1/4$  distinguishes the cases. Finally, exact knowledge of  $\lambda(C_\varphi)$  gives

$$S = \frac{2^N \lambda(C_\varphi) - 1}{2^{m-1}},$$

which is a polynomial-time metric reduction from  $\#\text{SAT}$ .  $\square$

## D Exact state reductions and local repair

This appendix proves the exact reductions, sharp examples, and computational boundaries collected in Section 4.

### Source-power reduction.

*Proof of Proposition 4.2.* Let  $C$  lift a function on  $\mathcal{C}$  to a function on  $\mathcal{X}$  by  $(Cf)(x) = f(\phi(x))$ . Condition (69) is exactly  $B_s C = C R_s$ . Since  $C\mathbf{1} = \mathbf{1}$ , iteration gives (70). For a nonnegative matrix  $M$ ,  $\|M\|_\infty = \max_x (M\mathbf{1})(x)$ . Surjectivity of  $\phi$  and (70) therefore give

$$\|B_s^T\|_\infty = \|R_s^T\|_\infty \quad \text{for every } T.$$

Taking  $T$ th roots proves (71) by Gelfand's formula. The eigenfunction statement follows again from  $B_s C = C R_s$ . Equation (72) has row sum one, and multiplication along a path telescopes to

$$Q_s[w] = P[w]^s \varrho_s^{-T} \frac{h_s(x_T)}{h_s(x_0)},$$

which is (73).  $\square$

### Extremal local repair.

*Proof of Theorem 4.3.* The classical Cartier–Foata identity gives

$$\sum_{h \in \mathcal{M}(D)} \prod_i z_i^{N_i(h)} = \frac{1}{P_D(\mathbf{z})} \tag{138}$$

as a formal power series (Cartier and Foata, 1969; Jerrum, 2024). For extremal partial rejection sampling, traces are in bijection with resampling transcripts: swapping two consecutive nonadjacent labels swaps repairs on disjoint scopes and changes neither the resampling table nor its frontier. The diamond property also makes the transcript independent of the rule used to choose among currently true events (Guo et al., 2019; Jerrum, 2024).

Fix a trace  $h$ . Its resampling blocks and final frontier partition the entries of the consumed resampling table. A halting table with transcript  $h$  is obtained by putting a configuration from  $\mathcal{B}_i$  in every block labelled  $i$  and a satisfying assignment in the final frontier. Conversely, every such choice realizes the transcript. One way to see the converse is to use the Cartier–Foata layers of  $h$  and read them backward. Writing a bad configuration into every scope of one layer makes exactly the events of that layer true: their scopes are disjoint; their neighbors are excluded by extremality;

and every nonneighbor is unchanged from the next layer. The Cartier–Foata condition says that every event in the next layer is a neighbor of, or equal to, an event in the current layer. Induction from the satisfying frontier proves the claim.

All cells in this partition are independent under the source. After taking the  $s$ th power and summing their values, a block labelled  $i$  contributes  $r_i(s)$  and the final frontier contributes  $Z_\Sigma(s)$ . Thus a trace  $h$  contributes

$$Z_\Sigma(s) \prod_i r_i(s)^{N_i(h)}.$$

Summing first over traces of length  $T$  and then over  $T$ , and applying (138), proves (77)–(78).

The same partition proves the upper bound in (79), with  $C_s = \sum_x \pi(x)^s$ : simply allow every full assignment at the current frontier. The lower bound must also respect the fixed sequential selector. Fix a satisfying assignment  $y$  and, for each  $i$ , a configuration  $b_i \in \mathcal{B}_i$ . Take any trace of length  $T$  and any choice of bad configurations in its blocks, put  $y$  in the final frontier, and run the fixed selector. This is a halting tape with that trace. Let  $i$  be the event repaired last. Every cell of the final frontier on  $V_i$  is drawn by this last repair. Replace just those newly drawn values, from  $y|_{V_i}$  to  $b_i$ . No earlier state or repair choice changes, while the state after the  $T$ th repair has exactly  $A_i$  true: its neighbors are excluded by extremality and its nonneighbors have disjoint scopes and remain false.

This map from the tables with fixed final frontier  $y$  to live prefixes is injective: the run identifies its last label  $i$ , after which restoring  $y|_{V_i}$  recovers the original table. It changes source probability by the factor  $\pi_{V_i}(b_i)/\pi_{V_i}(y|_{V_i})$ . Consequently the lower bound holds with

$$c_s = \pi(y)^s \min_i \left( \frac{\pi_{V_i}(b_i)}{\pi_{V_i}(y|_{V_i})} \right)^s > 0.$$

This proves (79) without assuming that the selector finishes one Cartier–Foata layer before starting the next.

By Cauchy–Hadamard and (79), the source-power growth of live prefixes is  $R(s)^{-1}$ . At  $s = 1$ , extremality and inclusion–exclusion give  $Z_\Sigma(1) = \mathbb{P}(\Sigma) = P_D(\mathbf{p}) > 0$ . More is true. For any  $\mathbf{y}$  with  $0 \leq y_i \leq p_i$ , independently retain event  $A_i$  with probability  $y_i/p_i$ . The retained events are still extremal, their marginals are  $\mathbf{y}$ , and their common avoidance event contains  $\Sigma$ . Inclusion–exclusion therefore gives

$$P_D(\mathbf{y}) \geq \mathbb{P}(\Sigma) > 0 \quad (0 \leq \mathbf{y} \leq \mathbf{p}).$$

Compactness and continuity then give  $P_D(u\mathbf{p}) > 0$  for all  $u \leq 1 + \varepsilon$ , for some  $\varepsilon > 0$ . Thus  $R(1) > 1$ . Summing the exact halting probabilities at  $s = 1$  gives (81). The first assertion of (80) follows; the existence of the exponential-rate limit follows from the standard finite Cartier–Foata automaton.

Every entry of every bad local configuration has source probability strictly between zero and one. Put  $c = \max_{i,b \in \mathcal{B}_i} \pi_{V_i}(b) < 1$ . If  $t > s$ , then  $r_i(t) \leq c^{t-s} r_i(s)$  for every  $i$ , and hence  $a_T(t) \leq c^{(t-s)T} a_T(s)$ . The finite Cartier–Foata automaton also shows that the growth  $R(s)^{-1}$  is continuous. It is therefore strictly decreasing, is less than one at  $s = 1$ , and is at least one at  $s = 0$  because the trace  $i^T$  exists for every  $T$ . This proves the existence and uniqueness of  $\delta$ .

For completeness, the same conclusion can be read directly from the full assignment chain for the fixed selection rule. Its total  $s$ -weight after  $T$  live repairs is  $L_T(s)$ ; the initial assignment contributes only a fixed finite starting vector. Decompose its finite nonterminal graph into reachable cyclic strongly connected components and apply Proposition 4.1 to each component. Equation (79) says that the largest component pressure is  $\log R(s)^{-1}$ , so the largest component root is  $\delta$ . This proves  $\dim_H^P \mathcal{N} = \delta$ . Every infinite path eventually remains in one of these finitely many components.

Under computability, coding in that component gives the upper effective-strong-dimension bound, and a Martin–Löf-random path for a critical component attains  $\delta$ .

For the pointwise statement, the finite component matrices have computable entries. The strict monotonicity just proved makes the critical root computable by bisection, and the normalized positive Perron vector of a critical irreducible component is computable. Hence its Doob law satisfies the effectivity hypothesis of Proposition 4.1.

Finally, the trace series has nonnegative parameters. Pringsheim’s theorem therefore implies that its first singularity on every positive ray occurs at a positive zero of  $P_D$ . At  $s = \delta$  that singularity is  $u = 1$ , which proves (82). Equivalently, the open domain of convergence of (138) consists of the vectors  $\mathbf{z}$  for which  $P_D(\mathbf{y}) > 0$  for every  $0 \leq \mathbf{y} \leq \mathbf{z}$ ; this is the standard independence-polynomial description of Shearer’s region (Shearer, 1985; Scott and Sokal, 2005).  $\square$

*Proof of Corollary 4.4.* For  $0 \leq s \leq 1$ , concavity gives, event by event,

$$p_i^s = \left( \sum_{b \in \mathcal{B}_i} \pi_{V_i}(b) \right)^s \leq r_i(s) \leq k_i^{1-s} p_i^s.$$

The trace parameters in (76) are coordinatewise increasing in these weights. Their exponential growth rates are strictly decreasing in  $s$ , so comparison at every  $s$  brackets their critical roots and proves (84). When the weights are constant across vertices, the trace growth is  $\kappa_D$  times that common weight. Solving the two scalar equations gives (85).  $\square$

*Proof of Corollary 4.5.* Atomicity gives  $r_i(s) = p_i^s$ . In the equal-probability case,  $a_T(s) = p^{sT} a_T(0)$ , so its exponential growth is  $p^s \kappa_D$ . Setting  $s = 1$  gives the survival rate and setting  $s = \delta$  makes the growth equal to one, proving (87). In general, write  $\alpha = -\log_2 \gamma(1)$ , where  $\gamma(s)$  is the trace growth at weights  $p_i^s$ . For  $0 \leq s \leq 1$ , each length- $T$  trace satisfies

$$2^{(1-s)\ell_{\min} T} \prod_i p_i^{N_i(h)} \leq \prod_i p_i^{sN_i(h)} \leq 2^{(1-s)\ell_{\max} T} \prod_i p_i^{N_i(h)}.$$

Taking exponential growth rates and using  $\gamma(\delta) = 1$  proves (88).  $\square$

### Hardness of the extremal critical power.

*Proof of Theorem 4.6.* The membership-hardness construction of Harvey et al. (2018) starts from a 4-regular graph  $G$ , adds a vertex  $\star$  adjacent to every vertex of  $G$ , and queries vectors

$$\mathbf{z} = (1/k, \dots, 1/k, z_\star), \quad |V(G)| < k < |V(G)|^2.$$

Deciding membership of these rational vectors in the open Shearer region is  $\#P$ -hard under polynomial-time Turing reductions. Queries with  $z_\star = 0$  are immediate. Write  $D = G + \star$ . Since the apex is adjacent to every vertex of  $G$ ,

$$P_D(1/k, \dots, 1/k, z_\star) = P_G(1/k, \dots, 1/k) - z_\star.$$

The vector  $(1/k, \dots, 1/k)$  lies in the open Shearer region of  $G$ . For any vertex  $v$ , deletion recurrence and monotonicity under deleting events give

$$P_G = P_{G-v} - \frac{1}{k} P_{G-N[v]} \leq (1 - 1/k) P_{G-v} \leq 1 - 1/k.$$

Thus queries with  $z_\star \geq 1 - 1/k$  are outside the open region. It therefore suffices to simulate queries satisfying

$$0 < z_\star < 1 - \frac{1}{k}. \quad (139)$$

Let  $D$  be the queried graph, let  $d_i$  be the degree of  $i$ , and put

$$d_{\max} = \max_i d_i, \quad z_{\max} = \max_i z_i,$$

$$L = \left\lceil \frac{d_{\max} + 2}{1 - z_{\max}} \right\rceil, \quad p_i = z_i^L.$$

Since  $-\ln z \geq 1 - z$  for  $0 < z < 1$ ,

$$p_i \leq e^{-(d_{\max}+2)} < 2^{-(d_i+2)}. \quad (140)$$

Orient every edge of  $D$ . Introduce an independent fair bit  $X_e$  for each edge and an independent private bit  $Y_i$  for each vertex, with

$$\mathbb{P}\{Y_i = 1\} = t_i = 2^{d_i} p_i \leq \frac{1}{4}.$$

For an oriented edge  $e = i \rightarrow j$ , prescribe 0 at endpoint  $i$  and 1 at endpoint  $j$ . Let  $A_i$  be the event that  $Y_i = 1$  and every incident edge bit has the value prescribed at  $i$ . It is atomic and

$$\mathbb{P}(A_i) = t_i 2^{-d_i} = p_i.$$

Adjacent events demand opposite values of their shared edge bit and are disjoint; nonadjacent events have disjoint scopes. The system is therefore extremal with dependency graph  $D$ . The assignment with every  $Y_i = 0$  is satisfying.

By atomicity,  $r_i(s) = p_i^s = z_i^{Ls}$ , so  $\mathbf{r}(1/L) = \mathbf{z}$ . Corollary 4.5 and downward closure of the open Shearer region give the required direction: because every  $p_i < 1$ , the vector  $\mathbf{p}^s$  decreases coordinatewise as  $s$  increases; it is on the boundary at  $s = \delta$  and lies in the open region exactly when  $s > \delta$ . Hence

$$\dim_H^P \mathcal{N} < \frac{1}{L} \iff \mathbf{z} \text{ lies in the open Shearer region of } D.$$

All constructed probabilities and the priority selector are computable, so Corollary 4.5 gives the same equivalence for maximal effective strong dimension.

Finally, (139) gives  $1 - z_{\max} \geq 1/k$  in the nontrivial queries, so  $L$  is polynomially bounded and is valid in unary. Raising a rational of polynomial bit length to the power  $L$  preserves polynomial description length. Each original vertex has degree five after the universal vertex is added, hence its event has scope six; only the universal-vertex event has linear scope. This completes the reduction.  $\square$

### The local termination boundary.

*Proof of Proposition 4.7.* Fix a flawless assignment  $x^*$ . At a live assignment  $x$ , the selected event  $A$  is present at  $x$  and absent at  $x^*$ . Hence  $x$  and  $x^*$  differ on at least one coordinate of  $v(A)$ . If the redraw of  $v(A)$  equals  $x_{v(A)}^*$ , the Hamming distance to  $x^*$  strictly decreases and no new disagreement is created. Its conditional probability is at least  $\beta$ , regardless of the past or of the selector. Each such redraw strictly decreases the Hamming distance to  $x^*$ . Consequently, after at most  $n$  favorable

live redraws no bad event can remain: if one did, its scope would contain a disagreement with  $x^*$  and another favorable redraw would decrease the distance again. Thus, from every live history, termination within the next  $n$  repairs has conditional probability at least  $\beta^n$ . Iteration proves (90). Conversely, without a flawless assignment there is always a present bad event, so the stopping condition is never reached.  $\square$

*Proof of Theorem 4.8.* Each repair has eight equiprobable redraws. If  $B_s$  is the live source-power matrix on flawed assignments, then entrywise counting gives

$$B_s = 8^{1-s} P_\Phi.$$

Every flawed assignment has a live self-loop: redraw the selected clause to its current values. Thus  $\lambda_\Phi > 0$  whenever the live state space is nonempty. Proposition 4.1, applied to the full-support initial law, now gives

$$1 = \rho(B_{\delta_\Phi}) = 8^{1-\delta_\Phi} \lambda_\Phi,$$

which is (91).

If  $\Phi$  is unsatisfiable, every assignment is flawed. Hence  $P_\Phi$  is stochastic,  $\lambda_\Phi = 1$ , every tape is nonterminating, and  $\delta_\Phi = 1$ .

Suppose instead that  $x^*$  satisfies  $\Phi$ . At any flawed assignment  $x$ , the selected clause is false at  $x$  and true at  $x^*$ . Its scope therefore contains a coordinate on which  $x$  and  $x^*$  differ. Redrawing all three coordinates to their values in  $x^*$  has probability  $1/8$ , strictly lowers the Hamming distance to  $x^*$ , and creates no new disagreement with  $x^*$ . After at most  $n$  consecutive favorable live redraws, the computation must terminate. Uniformly over the initial flawed assignment, the probability of this event is at least  $8^{-n}$ , and consequently

$$P_\Phi^n \mathbf{1} \leq (1 - 8^{-n}) \mathbf{1}, \quad \lambda_\Phi \leq (1 - 8^{-n})^{1/n} < 1.$$

This proves (92).

Exact 3-SAT remains NP-complete when every variable occurs in at most four clauses (Tovey, 1984). In such a formula a clause shares a variable with at most  $3(4 - 1) = 9$  other clauses. A satisfying assignment can be checked in polynomial time and, through the equivalence above, proves both strict inequalities. This establishes NP-completeness; complementing gives the coNP statements.  $\square$

### Disagreement repair.

*Proof of Theorem 4.9.* Let  $M(x) = \sum_{v \in V} x_v$ . Every selected edge contains one zero and one one. If its redraw contains  $J$  ones, then

$$M_{t+1} = M_t - 1 + J, \quad J \sim \text{Binomial}(2, p). \quad (141)$$

Because  $G$  is connected, an assignment is live exactly when  $1 \leq M \leq N - 1$ . Conditional on any past with  $M_t = k$ , the four possible redraws have  $s$ -weights  $q^{2s}, (pq)^s, (pq)^s, p^{2s}$  and move the count by  $-1, 0, 0, +1$ , respectively. Induction on  $T$  therefore shows directly that the source-power sums grouped by  $M$  evolve by (94), even when the selector uses the past. The initial value is  $\text{Binomial}(N, p)$ , proving (95) and (96). This also proves that the full absorption-time law is independent of  $G$  and of the selection rule.

For  $1 \leq k < N$ , the positive function  $h_s$  in (103) satisfies

$$\begin{aligned}(R_s h_s)(k) &= q^{2s} h_s(k-1) + 2(pq)^s h_s(k) + p^{2s} h_s(k+1) \\ &= 4(pq)^s c_N^2 h_s(k).\end{aligned}$$

Proposition 4.2 therefore gives (97). Equivalently,  $R_s$  is diagonally similar to  $(pq)^s$  times the tridiagonal matrix with diagonal 2 and off-diagonal 1; its eigenvalues are

$$4(pq)^s \cos^2\left(\frac{r\pi}{2N}\right), \quad r = 1, \dots, N-1. \quad (142)$$

Equations (98) and (99) follow by setting  $s = 1$  and by solving  $\varrho_s = 1$ , respectively.

Applying (72) with  $h_s$  gives

$$\frac{P_p(z)^s h_s(k + d(z))}{\varrho_s h_s(k)} = \frac{\sin((k + d(z))\vartheta)}{4c_N^2 \sin(k\vartheta)}.$$

The right-hand side is independent of  $s$  and  $p$ , and its sum over the four redraws is one because

$$\sin((k-1)\vartheta) + 2\sin(k\vartheta) + \sin((k+1)\vartheta) = 4c_N^2 \sin(k\vartheta).$$

Moreover,  $\nu^*(k)Q^*(11 \mid k) = \nu^*(k+1)Q^*(00 \mid k+1)$ ; the self-loops need no check. This proves reversibility and (101). (102) is (73). At the root  $s = \delta_{N,p}$ , endpoint ratios are bounded above and below because  $1 \leq k < N$ . Equation (96) gives the Hausdorff upper bound directly. The global positive law  $Q^*$  and the bounded likelihood ratio give the matching mass-distribution lower bound and, under computability, the effective-dimension upper bound and attainment on a  $Q^*$ -Martin-Löf-random tape. This argument does not require the full assignment chain for a particular selector to be irreducible.

When  $p$  and the selection rule are computable, the explicit formula makes  $\delta_{N,p}$  computable, and the entries of  $Q^*$  are computable algebraic numbers. Finally, suppose  $p = 1/2$  and let  $u_k = \mathbb{E}[\tau \mid M_0 = k]$ . First-step analysis gives

$$u_0 = u_N = 0, \quad u_k = 1 + \frac{1}{4}u_{k-1} + \frac{1}{2}u_k + \frac{1}{4}u_{k+1}, \quad 1 \leq k < N.$$

The unique solution is  $u_k = 2k(N-k)$ . Since  $M_0 \sim \text{Binomial}(N, 1/2)$ ,

$$\mathbb{E}[M_0(N - M_0)] = \frac{N(N-1)}{4},$$

and averaging  $u_{M_0}$  proves (108). □

*Proof of Proposition 4.10.* For each  $S \subseteq \{1, \dots, N-1\}$ , let  $x_0^S = 0$  and let  $x_i^S = 1$  exactly when  $i \in S$ . Give every repair the common future redraw 00. The successive repair labels are the elements of  $S$  in increasing order, after which the computation halts. Distinct sets therefore have distinct output sequences under the same continuation. If two of the  $x^S$  had the same summary, deterministic prediction and updating would give the same output sequence, a contradiction. □

**Adjacent rises.** Let  $n \geq 1$  and let  $X_0, \dots, X_n$  be independent bits with  $\mathbb{P}\{X_i = 1\} = p_i \in (0, 1)$ , and set

$$A_i = \{X_{i-1} = 0, X_i = 1\}, \quad 1 \leq i \leq n.$$

At each step, any deterministic current-assignment rule may select a true  $A_i$ ; the repair redraws  $(X_{i-1}, X_i)$  from its product marginal. For  $\theta \geq 0$ , let  $P$  be the sequential source law and let  $B_\theta$  be the full live-state source-power operator. For a pair outcome  $z = (z_{i-1}, z_i)$ , write

$$P_i(z) = (1 - p_{i-1})^{1-z_{i-1}} p_{i-1}^{z_{i-1}} (1 - p_i)^{1-z_i} p_i^{z_i}.$$

Define the symmetric tridiagonal matrix  $K_\theta$ , indexed by  $0, \dots, n$ , by

$$(K_\theta)_{i-1,i} = (K_\theta)_{i,i-1} = [(1 - p_{i-1})p_i]^\theta, \quad (K_\theta)_{ii} = 0. \quad (143)$$

For the state lower bound below, an *outcome-driven state representation* is a map  $\sigma$  from assignments to a finite set together with a status map and an update map  $U$  such that the state alone reports live versus halted and

$$\sigma(F(x, z)) = U(\sigma(x), z)$$

whenever the deterministic rule repairs the live assignment  $x$  and observes the pair outcome  $z \in \{00, 01, 10, 11\}$ ; here  $F(x, z)$  is the resulting assignment. Thus no part of the current assignment is available outside the state.

**Theorem D.1** (Adjacent-rise pressure). *For every deterministic current-assignment selection rule and every  $\theta \geq 0$ ,*

$$\rho(B_\theta) = \rho(K_\theta)^2. \quad (144)$$

Consequently,

$$\lim_{T \rightarrow \infty} \mathbb{P}\{\tau \geq T\}^{1/T} = \rho(K_1)^2, \quad \dim_H^P \mathcal{N} = \delta, \quad \rho(K_\delta) = 1. \quad (145)$$

The root  $\delta \in [0, 1]$  is unique. If the biases and selection rule are computable, it is also the largest effective strong dimension of a nonterminating tape and is attained by an explicit edgewise product-form law.

For the least-index rule, however, every outcome-driven state representation has at least  $2^n$  states, including the halting state.

**Corollary D.2** (A linear eigenproblem versus exponential continuation state). *For fixed  $\theta$ , the matrix  $K_\theta$  has  $2n$  nonzero entries and is constructed in linear time. Its largest eigenvalue, and hence  $\rho(B_\theta)$ , can be approximated to additive error  $\varepsilon$  using  $O(n \log(1/\varepsilon))$  arithmetic operations by Sturm bisection for a symmetric tridiagonal matrix. The dimension  $\delta$  is then obtained by one-dimensional bisection in  $\theta$ . This is an arithmetic-operation statement; bit complexity also depends on how the biases are represented and on numerical conditioning. In contrast, the least-index process has  $2^n$  outcome-driven continuation states.*

Adjacent events  $A_i$  and  $A_{i+1}$  are mutually exclusive, while nonadjacent events are independent. Thus this is an extremal instance in the terminology of partial rejection sampling: dependent bad events cannot occur simultaneously (Guo et al., 2019; Jerrum, 2024). In this regime, known Cartier–Foata enumeration identifies repair transcripts with traces and gives the generating function  $P_D(\mathbb{P}(A_1)z_1, \dots, \mathbb{P}(A_n)z_n)^{-1}$ , where  $P_D$  is the signed independence polynomial of the dependency

graph. The theorem above uses the same structure at every source power and makes its tape-geometric consequence explicit; it does not claim a new conditional-output theorem or a new trace enumeration.

For the dependency path  $D$ , put

$$q_\theta = P_D(\mathbb{P}(A_1)^\theta, \dots, \mathbb{P}(A_n)^\theta) = \sum_{I \in \mathcal{I}(D)} (-1)^{|I|} \prod_{i \in I} \mathbb{P}(A_i)^\theta.$$

The tridiagonal determinant recurrence gives  $q_\theta = \det(I - K_\theta)$ . At  $\theta = \delta$  the full determinant vanishes; deleting any event strictly lowers the Perron root, so every proper subfamily still has positive Shearer polynomial. Hence the powered marginal vector reaches the Shearer boundary exactly at  $\theta = \delta$  (Shearer, 1985; Scott and Sokal, 2005). The determinant and boundary description are classical. The conclusion here is that this boundary value is also the source-relative dimension of the nonterminating tapes of the sequential resampling computation.

### Proofs for adjacent rises.

*Proof of Theorem D.1.* For a live word  $x = x_0 \cdots x_n$ , let  $a(x)$  be its first zero and  $b(x)$  its last one. Then  $a(x) < b(x)$ , and the sets

$$C_{a,b} = \{1^a 0 u 1 0^{n-b} : u \in \{0, 1\}^{b-a-1}\}, \quad 0 \leq a < b \leq n,$$

partition the live assignments. Every repair can only increase  $a$  or decrease  $b$ . Hence  $B_\theta$  is block upper triangular over the intervals  $[a, b]$ .

Fix one diagonal block and write

$$\alpha_j = (1 - p_j)^\theta, \quad \beta_j = p_j^\theta, \quad c_i = \sqrt{\alpha_{i-1} \beta_i}.$$

Let  $\eta$  and  $v = (v_a, \dots, v_b) > 0$  be the Perron value and vector of the weighted path  $K_{\theta, [a, b]}$ . For  $a < j < b$ , put

$$t_j = \sqrt{\frac{\alpha_{j-1} \alpha_j}{\beta_j \beta_{j+1}}} \frac{v_{j-1}}{v_{j+1}}, \quad h(x) = \prod_{j=a+1}^{b-1} t_j^{x_j}. \quad (146)$$

Set  $t_a = 0$  and  $t_b = +\infty$  for the boundary formulas. The Perron recurrence gives, for  $a \leq j < b$ ,

$$U_j := \alpha_j + \beta_j t_j = \eta \sqrt{\frac{\alpha_j}{\beta_{j+1}}} \frac{v_j}{v_{j+1}}, \quad t_i = \frac{U_{i-1} U_i}{\eta^2} \quad (a < i < b). \quad (147)$$

If the selected rise is  $i$ , summing  $P_i(z)^\theta h(y)/h(x)$  over the outcomes that retain the endpoints of  $[a, b]$  gives

$$\frac{(B_{\theta, [a, b]} h)(x)}{h(x)} = \begin{cases} \alpha_a (\alpha_{a+1}/t_{a+1} + \beta_{a+1}), & i = a + 1, \\ U_{i-1} (\alpha_i/t_i + \beta_i), & a + 1 < i < b, \\ \beta_b U_{b-1}, & i = b. \end{cases}$$

Equations (147) and the two endpoint Perron equations make each expression equal to  $\eta^2$ . The same is immediate when  $b = a + 1$ . Thus  $B_{\theta, [a, b]} h = \eta^2 h$ , independently of which true rise the rule selects. Since  $h > 0$ , Collatz–Wielandt gives

$$\rho(B_{\theta, [a, b]}) = \rho(K_{\theta, [a, b]})^2.$$

Every proper interval matrix is a proper principal submatrix of the irreducible  $K_\theta$ , so strict Perron comparison makes  $[0, n]$  the unique maximal block. This proves (144); the finite-state pressure formula gives (145). For  $n \geq 2$ ,  $\rho(K_0) = 2 \cos(\pi/(n+2)) > 1$ . On the other hand, from every live word a sequence of at most  $n+1$  outcomes 00 removes the right endpoint of one selected rise at each step and reaches a terminal nonincreasing word  $1^a 0^{n+1-a}$ . The probability of this sequence is bounded below uniformly over the finite state space. Hence  $\rho(B_1) < 1$  and therefore  $\rho(K_1) < 1$ . Strict monotonicity of  $\rho(K_\theta)$  now gives a unique root in  $(0, 1)$ . For  $n = 1$ ,  $\rho(K_0) = 1$  and  $\delta = 0$ .

The law attaining the dimension is also local. At  $\theta = \delta$ , use the preceding  $t_j$  on  $C_{0,n}$  and define

$$U_j = \alpha_j + \beta_j t_j, \quad V_j = \alpha_j / t_j + \beta_j, \quad U_{i-1} V_i = 1.$$

Here  $t_0 = 0$  and  $t_n = +\infty$ , interpreted by limits. Thus  $U_j$  is used for  $0 \leq j < n$  and  $V_j$  for  $0 < j \leq n$ . If rise  $i$  is selected, redraw its two coordinates independently according to

$$\begin{aligned} L_j(0) &= \alpha_j / U_j, & L_j(1) &= \beta_j t_j / U_j, \\ R_j(0) &= \alpha_j / (t_j V_j), & R_j(1) &= \beta_j / V_j, \end{aligned}$$

using  $L_{i-1} \otimes R_i$ . The boundary marginals are degenerate, so the run remains in  $C_{0,n}$ . Direct substitution shows that this product law is

$$Q_\delta(z \mid x) = P_i(z)^\delta \frac{h(y)}{h(x)}. \quad (148)$$

Thus along every surviving prefix  $w$  in this component,

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h(x_T)}{h(x_0)}. \quad (149)$$

The endpoint term is bounded. The mass-distribution argument proves attainment and the Hausdorff lower bound. For the upper and pointwise bounds, note that along any infinite run the pair  $(a, b)$  changes at most  $2n$  times and is eventually fixed. Applying the same componentwise coding argument on the final  $C_{a,b}$ , whose pressure root is at most  $\delta$ , proves the effective and Hausdorff upper bounds for all nonterminating tapes. When the biases are computable, the finite tridiagonal matrix has computable entries, strict monotonicity makes  $\delta$  computable by bisection, and its normalized Perron vector is computable. Thus the displayed critical law is computable, as required for the pointwise attainment statement.

It remains to prove the state bound. For each  $S \subseteq \{1, \dots, n\}$ , let  $x_0^S = 0$  and  $x_i^S = 1$  exactly for  $i \in S$ . Under the least-index rule, repeated outcome 00 deletes the leftmost one; this distinguishes sets of different sizes. If  $S = \{s_1 < \dots < s_k\}$  and  $T = \{t_1 < \dots < t_k\}$  are distinct, choose  $r < k$  with unequal suffix sums. After  $r$  outcomes 00, repeated outcome 10 decreases by one the inversion count

$$I(x) = \#\{(i, j) : i < j, x_i = 0, x_j = 1\}.$$

The two counts are

$$\sum_{j=r+1}^k s_j - \binom{k-r}{2} \quad \text{and} \quad \sum_{j=r+1}^k t_j - \binom{k-r}{2},$$

so one run halts first under a common continuation. The  $2^n$  assignments  $x^S$  are therefore pairwise continuation-distinct.  $\square$

*Proof of Corollary D.2.* The characteristic polynomials of the leading principal submatrices satisfy the three-term recurrence

$$D_{j+1}(\lambda) = \lambda D_j(\lambda) - (K_\theta)_{j-1,j}^2 D_{j-1}(\lambda).$$

It costs  $O(n)$  operations to evaluate the Sturm sequence at a trial  $\lambda$ , and  $O(\log(1/\varepsilon))$  trials isolate the largest root in a fixed bounded interval. The remaining claims follow from Theorem D.1 and the strict monotonicity of  $\rho(K_\theta)$ .  $\square$

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